

What Do Good Managers Do and What Do They Leave Behind?*

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Abstract

Using administrative data from a large Colombian retailer and exploiting a manager rotation policy to estimate manager value-added, we study what good managers do and whether their impacts persist after they leave. Good managers increase sales by 12%, reducing stockouts while making inventory leaner, generating larger revenue gains from price changes without increasing the frequency of repricing, and shifting personnel attention toward back-office operations. When they depart, the staffing and inventory improvements persist, but pricing-related gains erode. These patterns reveal a fundamental distinction: some managerial contributions are embedded in organizational structure, while others depend on the manager’s ongoing, context-specific judgment.

Keywords: managerial attention, span of control, allocative skill, managerial quality, retail
JEL: J24, M10, M53

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1 Introduction

Managerial decision-making is fundamentally a problem of allocating scarce resources: the manager’s own time and attention; their teams’ efforts; and financial and operational capacity across the multitude of tasks that constitute the production function (Adhvaryu et al., 2022; Benson and Shaw, 2025; Weidmann et al., 2026). This decision-making can be staggeringly complex. *The Wall Street Journal*, for example, recently chronicled a day in the life of a Walmart store manager in Texas (Nassauer, 2024). The manager was responsible for 300 employees and over \$100 million in annual sales. In a single workday, she handled negative customer feedback, staffing shortfalls, inventory issues, merchandising adjustments, coaching subordinate managers, and the coordination of high-value online-order fulfillment. Some of what she accomplished turned on her own real-time judgment – deciding, for instance, to stock eclipse-themed gear based on her read of local demand in the lead-up to a solar eclipse. Other tasks she managed by drawing on her team and the organizational routines around her.

What do successful managers do differently from others with respect to these decisions? To what extent are the gains from good management embedded in organizational processes or personnel, and to what extent do they depend on the manager’s continued presence and inputs? In other words, do managers leave their imprint on the organization, and if so, what behavior and outcomes actually persist? These questions are central to the understanding of organizations, but answering them has proven difficult. Manager effects have been causally identified in many settings using exogenous manager “moves” in two-way fixed effects frameworks.¹ But identifying what, specifically, high-performing managers do differently – and whether the gains they generate persist after they leave – has proven difficult.

Our study aims to overcome this challenge. We use rich administrative data from a multi-billion-dollar Colombian retailer, including scanner-level sales data, detailed inventory records, supplier characteristics, and supplier-store transactions. We exploit the firm’s policy of rotating managers across stores under plausibly exogenous assignment between January 2017 and June 2020 to separately identify manager and store fixed effects on productivity in a framework similar to Abowd et al. (1999). Using these estimates, we classify high-type (low-type) managers as those whose estimated fixed effects fall above (below) the median of the manager fixed effect distribution, and high- (low-) manager-quality stores as those whose average manager fixed effect falls above (below) the median across stores.

We document three key facts. First, good managers improve inventory outcomes.

¹See, e.g., Abowd et al. (1999); Adhvaryu et al. (2024a); Caplin et al. (2023); Hoffman and Stanton (2024); Lazear et al. (2015); Metcalfe et al. (2023); Weidmann et al. (2026)

Stockouts decrease by 11.5 percent, directly enabling sales by ensuring product availability.² At the same time, good managers make inventory leaner, with sizable reductions in both the overall dollar value of inventory and the ratio of inventory to sales. The combination of fewer stockouts and leaner inventories is consistent with more effective inventory adjustment, rather than a simple expansion of inventory across products to avoid stockouts.

Second, good managers make more effective pricing decisions, as evidenced by larger revenue gains following price changes. Importantly, the overall frequency of price changes does not change. Instead, both price increases and price decreases implemented under good managers are more likely to generate revenue gains than were price changes prior to their arrival. This pattern is consistent with more context-appropriate pricing adjustments in response to local demand conditions. While sales also rise for products whose prices are not adjusted, the gains among products whose prices are changed (whether upward or downward) are nearly three times as large.

Third, good managers rebalance personnel across functions in ways that are closely conceptually aligned with these operational improvements. Upon the arrival of a good manager, we observe a significant increase in personnel allocated to back-office, non-customer-facing roles related to pricing, inventory, and supply chain management. In particular, good managers reduce turnover among back-office managers while increasing hiring into these departments. Critically, we do not observe a statistically significant increase in total staffing overall. Together, these operational changes account for the topline performance gain we estimate: store sales rise by nearly 12 percent within a year of a high-value-added manager’s arrival, with comparable gains in productivity, defined here as sales per worker.

We then combine these estimates with personnel records and a managerial survey capturing managerial style, traits, and priorities to characterize which types of managers are associated with superior performance. High-type managers do not exhibit higher cognitive or socio-emotional skills; instead, they report prioritizing communication and relationships with their teams. Consistent with the operational margins we document, they also report spending more time on inventory-related procedures and are more likely to identify pricing as a critical determinant of store performance.

Finally, we study what happens when a good manager leaves. To do so, we construct a similar event study around managerial departures rather than arrivals. We find that the gains associated with good managers persist only partially. Roughly a third of the sales and productivity gains realized after the arrival of a good manager are preserved after that manager leaves, with the rebalancing of staff toward back-office operations and the reduction in stockouts both persisting. By contrast, gains in the revenue yield of

²See [Conlon and Mortimer \(2013\)](#), who document large effects of stockouts on profitability.

price adjustments erode substantially following the manager’s departure. It is precisely among products with price changes that the reversal of the overall sales gains appears concentrated, while sales among products without price changes remain unchanged after departure. This pattern suggests that some components of managerial value-added persist through organizational reallocation and the operational outcomes it enables, while others depend more directly on the manager’s continued presence and ongoing, context-specific decision-making – most clearly, repricing decisions in response to evolving local demand.

1.1 Related Literature and Contributions

Three papers are closest to this study. [Metcalfe et al. \(2023\)](#) estimates manager effects on store productivity in a retail setting using manager moves across stores. [Weidmann et al. \(2026\)](#) uses a similar rotation-based identification strategy to characterize what predicts managerial value-added. [Adhvaryu et al. \(2023\)](#) identifies manager fixed effects in a manufacturing setting and documents that high-quality managers raise productivity. Together these papers establish, across a range of settings and with compelling identification strategies, that manager quality causally shapes firm performance. Yet identifying what, specifically, high-performing managers do differently ([Bandiera et al., 2020](#); [Bloom et al., 2013](#); [Friebel et al., 2022](#); [Gosnell et al., 2020](#); [Hoffman et al., 2018](#); [Hoffman and Stanton, 2024](#); [Lazear et al., 2015](#)) – and whether the gains they generate persist after they leave ([Bloom et al., 2020](#); [Bruhn et al., 2018](#)) – has proven difficult.

We endeavor to make progress on both margins. Using rich linked administrative, scanner-level, and supplier-store transactional data from a large Colombian retailer, we observe the operational decisions through which managerial value-added is realized: inventory management, pricing, and personnel allocation. We document that good managers reduce stockouts while making inventory leaner, generate larger revenue gains from price changes without increasing repricing frequency, and shift personnel attention toward back-office operations. Vision into managerial decision-making at this granular level allows us to move beyond composite quality measures and bring into focus what good managers do differently.

The natural experiment around managerial departures is a second contribution. By constructing event studies around both the arrivals and departures of high-value-added managers, we show that the persistence of managerial gains depends on their nature. Contributions embedded in organizational structure – back-office staffing changes and the inventory improvements they enable – outlast the manager. On the other hand, those requiring ongoing, context-specific judgment – e.g., pricing decisions in response to evolving local demand – erode when the manager leaves, with the overall sales reversal concentrated entirely among repriced goods. This distinction between what organizations

retain and what travels with the manager is, to our knowledge, novel in the literature.

A useful way to situate these findings is through recent work synthesizing how managers add value (Benson and Shaw, 2025; Hoffman and Tadelis, 2021). That work emphasizes six core managerial activities: hiring, retaining, training, motivating, evaluating, and allocating workers (Benson and Shaw, 2025). Our results reinforce the importance of worker allocation across tasks (Smeets et al., 2019) but also point to an additional dimension: the ability to adapt in this case specifically the ongoing exercise of context-specific judgment in response to evolving operational conditions. Our findings complement recent evidence that skillful allocation of constrained resources predicts manager value-added (Weidmann et al., 2026), and are consistent with evidence from service-sector settings where managers create value by adapting to changes in the business environment (Adhvaryu et al., 2026, 2024b).

This perspective helps clarify how our study relates to recent work on the long-run effects of managerial training and intervention (Bianchi and Giorcelli, 2022; Bloom et al., 2020; Bruhn et al., 2018; Giorcelli, 2019). In those settings, the intervention is built around practices that, by construction, must be codifiable in order to be teachable. Our setting allows a different exercise. Rather than beginning with training in a predefined bundle of best practices, we begin with the exogenous arrival of a high-value-added manager; observe directly in rich administrative data what such managers do differently in the course of normal operations; and trace which practices and gains are preserved after they leave. This makes it possible to study not only how much of managerial value-added persists, but which dimensions depend on the manager’s continued presence, and which are therefore harder to codify.

2 Context

Our partner firm operates one of the largest supermarket chains in South America. We leverage administrative data from 220 of the firm’s units between January 2017 and June 2020.

2.1 Store operations

The setup, layout, and operational structure of each store is highly standardized and centrally determined by corporate headquarters. The organizational structure has three layers, comprising central headquarters, middle management working in the stores (these managers supervise production activities and manage the day-to-day operations of each store), and front-line workers (who conduct all operating activities). Importantly, each

store’s data systems are integrated and standardized by our partner firm, ensuring that the metrics we rely on are standard across stores.

2.2 Employees and Sections of the Store

The typical store operated by our partner firm has 103 employees, works with an average of 826 suppliers, and sells 54,875 distinct products. As in many large supermarkets (e.g., Walmart, Target, or Costco), the type of products sold varies, with some stores selling everything from non-perishable durables like flat-screen TVs and car batteries to groceries. While many of these products are standard across stores, demand varies greatly store-to-store.

Within a store employees are spread across several departments (e.g., Bakery, Butchery, Cashiers, Customer Service, Product/Inventory Management, etc.). We refer to workers and managers whose main responsibilities are pricing- or inventory-related as “back-office” employees. Any worker or manager whose job title is not primarily associated with a pricing or inventory task is a “customer-facing” employee.³

This distinction is an important one: back-office workers and managers are focused on the logistics of stocking a large store (e.g., receiving shipments in the warehouse, physically stocking shelves, negotiating with suppliers, setting prices, etc.), while managers of the remaining teams tend to focus more on customer-facing tasks (e.g., ensuring speedy checkouts, managing customer service and returns, overseeing the deli, meats, and/or bakery, etc.).⁴ Of the 103 employees in an average store, 25 work in a back-office role and 77 in a customer-facing role (see Table 1). Stores hire an average of 4.73 workers per month, with 4.34 hires going to customer-facing jobs, and only 0.40 hires working in the back-office.

2.3 Managers

Managers can have one of four titles: “Director” is the highest position. This is followed by “Boss”, then “Leader”, and finally “Coordinator.” Table A.8 shows that Coordinators and Leaders are far more common than Bosses or Directors. The responsibilities and actions of each category of manager vary by store and over time, such that the distinction between the day-to-day actions of managers with different titles is not clear-cut. The clearest

³For example, stock control or warehouse workers, supply and sales coordinators, and price analysts are all back-office roles, while the store manager, cashiers, call center consultants, customer management, or butchery/bakery roles would all be customer-facing roles.

⁴While the complexities of personnel management and customer-facing work are not negligible (?), we find it useful to differentiate between the types of problems inherent to these roles and the types of problems that come from managing the logistics of keeping 55,000 products well-priced and appropriately stocked.

Table 1: Pre-treatment summary statistics for all stores

	All stores		
	Mean	Median	S.D.
Total workforce	102.94	81	101.37
Back-office workforce	25.36	19	27.29
Customer-facing workforce	77.58	60	75.57
Number of managers	8.01	6	7.13
Directors	0.68	1	0.72
Bosses	0.59	0	1.19
Leaders	4.66	1	6.65
Coordinators	2.08	0	2.76
Back-office managers	6.09	5	5.48
Customer-facing managers	1.92	1	2.67
Ratio managers to non-managers	0.11	0	0.10
Ratio managers to non-managers - Back-office only	0.48	0	0.55
Ratio managers to non-managers - Customer-facing only	0.05	0	0.07
Manager turnover	0.20	0	1.05
Back-office manager turnover	0.15	0	0.83
Customer-facing manager turnover	0.05	0	0.30
Total hiring	4.73	2	9.88
Back-office hiring	0.40	0	1.07
Customer-facing hiring	3.87	1	6.57
Hiring of managers	0.01	0	0.12
Hiring of Back-office managers	0.01	0	0.09
Hiring of Customer-facing managers	0.00	0	0.08
Within-store manager moves	0.02	0	0.22
Back-office manager moves (within-store)	0.02	0	0.20
Customer-facing manager moves (within-store)	0.01	0	0.08
Arrival of any manager (across store moves)	0.14	0	0.35

Table 1 reports means and standard deviations for organizational variables for all stores prior to the arrival of a good manager. All statistics are reported at the store-monthly level. *Total workforce* includes any person employed in the retail store during the period analyzed. *Managers* are defined as those workers who are coordinators, leaders, bosses and directors. Conversely, *Non-managers* are those workers who are not in those positions. *Turnover* indicates the number of employees in the store that left the company entirely (i.e., it excludes moves between stores), while *Hiring* represents the number of employees in the store that were hired. Finally, *Moves* indicates the number of moves from one position to another within the company (including vertical or horizontal moves).

distinction between manager types is that the lowest level of managers (coordinators) tend to be front-line managers of specialized departments (e.g., a shift manager for the deli section), while higher level managers are more likely to be the ones making store-wide decisions.

One distinction that is important in our context is that of back-office and customer-facing managers, with back-office managers generally sharing the duties relating to the logistics of stocking a large store (e.g., most or all back-office managers spend some time negotiating with suppliers, ordering goods, and setting prices). The lower panel of Table 1 presents statistics specific to the management team within stores. On average, stores employ 8 managers, 6.1 of whom are back-office managers. Back-office departments have a much narrower span of control than customer-facing departments: the ratio of managers to non-managers in back-office departments is 0.50, compared with 0.06 in customer-facing departments. On average, 0.16 managers arrive at a store in a given month. We discuss the movement of managers across stores at length in section 4. Manager turnover and hiring are both low, with the average store losing 0.20 managers to turnover, and gaining 0.01 managers to hiring each month.

2.4 What managers do

In addition to personnel-related tasks (e.g., conflicts between workers, training, scheduling), managers oversee the ordering and stocking of tens of thousands of products. Much of this product management is centralized, meaning that a store manager is free to depend on central headquarters to make decisions about when, how much, or from which suppliers to order. Despite this, demand for consumer goods varies by region (e.g., a winter coat will sell better in Medellín than Cali) meaning that each store needs to stock different products. And, even within a store, demand is sensitive to details like weather, time of year, and day of the month. This means that local managers often spend a significant amount of time adjusting when, how much, and from which suppliers to order.

Accordingly, maximizing store sales requires managers to optimize along two competing dimensions: (1) how many and which goods to monitor and manage closely; and then for each of these goods (2) how much to stock, when to restock and what prices to set or adjust. Specifically, allocating attention to any one product increases the chances that sales are maximized on that product, but simultaneously decreases the likelihood that the manager will maximize sales of all other products in the store. Conversely, spending too little attention on many products increases the chances of making an inventory or pricing error, even as it reduces the share of products being ignored. As the number of products (and/or suppliers) increases, so too does the likelihood that attention constraints bind.

2.5 Manager Rotation

During our period of analysis (January 2017 to June 2020), 28% of managers moved stores at least once. This mobility owes to our partner firm’s policy of periodically rotating

managers across stores. These moves are initiated by central headquarters. The stated rationale for moving managers is that rotating managers learn how other stores operate, a key part of their career development. We discuss the quasi-random nature of these movements in detail in section 4.1.

3 Data and Descriptive Statistics

We leverage two sources of data: primarily, granular administrative data collected by the supermarket chain supplemented by a survey we designed and administered to managers at the supermarket chain.

3.1 Administrative Data

We use administrative data from 220 of the supermarket chain’s stores that operated between January 2017 and June 2020. This data includes start and end dates, wages, promotions, and attendance logs for every employee. We also have sales, pricing, and inventory for every product as well as a record of all suppliers for any product sold. Each store’s data systems are integrated and standardized by our partner firm, ensuring that the metrics we rely on are standard across stores.

The most important indicator of store performance is sales. We also define productivity as a store’s sales divided by its number of employees.⁵ Two variables that are particularly important in the retail context are “stockouts” and value of inventory.

A *stockout* refers to the situation in which a store runs out of a given product due to excess demand.⁶ Minimizing the frequency of stockouts is a key task for back-office managers, both because a stockout means that the store is actually losing out on sales every time a customer would have purchased that product⁷, and because of potentially negative impacts on customer satisfaction.

⁵Our data does not include a reliable measure of the costs incurred when a store acquires a product from a supplier. Thus, we focus on sales instead of profits.

⁶We categorize cases in which a product sells out but is not re-purchased (i.e., the product is discontinued) as stockouts, even though this means we may slightly overcount unintentional stockouts. We do so due to data limitations—because we do not observe sales or inventory after the end of our sample we cannot say whether a manager has not yet had a chance to reorder the product or whether it was truly discontinued. Our inability to differentiate between truly discontinued products and temporary stockouts should result in attenuation of the stockout coefficient so long as treatment (1) decreases stockouts and (2) increases the number of discontinued products. Both assumptions seem plausible given the results presented below. Although we do not have a direct way to test the second assumption, a consistent interpretation of the complete set of actions taken by good managers suggests that if the number of discontinued products does, in fact, change under a good manager, it is more likely to increase than decrease.

⁷Of course, some within-store substitution of products is likely to occur, but on average, stockouts certainly reduce a store’s sales.

Stockouts can be avoided given a large enough inventory, but, of course, large inventories can have significant monetary-, time- and space-related costs. While it is unclear whether store managers fully internalize the cost of tying up money in inventory costs (i.e., it is unlikely that holding too many products in inventory means the store managers cannot afford to buy more products), it is very likely that managers experience time- and space-related costs of holding too much inventory.

For example, carrying excess inventory of perishable goods—or even non-perishable, seasonal products—may require stores to sell these products at lower prices to make room for newer products. Moreover, space constraints bind not only on customer-facing shelves, but also in the store’s warehouse: indeed, as seasons change a manager may prefer to sell larger non-perishable products (e.g., synthetic Christmas trees) for a loss to avoid storing them for several months, taking up space that could be utilized to meet higher demand products and increasing the likelihood of stockouts of those products and lost sales as a result.

Finally, a key input in determining both the frequency of stockouts and the value of inventory is how managers price their products. Table 2 reports summary statistics for several variables relating to stockouts, inventory size/value, and product pricing. On average, stores sell \$1.72 million USD per month, while holding \$479 thousand USD in inventory. In general, we observe that stores vary widely in sales and productivity (which we calculate as total sales per worker).

3.1.1 Survey of Managers

In 2023, we surveyed managers at all of our partner firm’s locations, achieving a 77.6% response rate. This survey captured information on managers’ demographics, job histories, psychometric profiles, personality traits, managerial practices, leadership styles, job descriptions and job satisfaction. This survey data has two limitations:

First, the survey was conducted nearly three years after the end of our administrative data sample.⁸ Since some managers from our 2017-2020 sample were not the same as the managers surveyed in 2023, we can analyze responses only for individuals who were managers in both time periods.⁹ Fortunately, because managerial turnover is low, and initial survey take-up was high we still have a workable sample size of 183 respondents. Second, this survey targeted only higher-ranking middle managers, while our empirical

⁸The administrative data is described in detail in section 3.

⁹Even though many managers rotated to different stores in the years between 2020 and 2023, we match the survey respondents back to the store(s) in which they worked during our 2017-2020 period of analysis. Thus, any mismatch between the samples is due to attrition between 2020 and 2023. While it’s unclear how differential attrition would affect the generalizability of our survey findings, we do note that attrition does not appear to be differential along the lines of manager type (i.e., we cannot reject the hypothesis that high- or low-type managers were equally likely to have responded to the 2023 survey ($p=0.29$)).

Table 2: Pre-treatment summary statistics for sales-, pricing- and inventory-related outcomes

	All stores		
	Mean	Median	S.D.
Sales (USD)	1,723,302.38	1,351,463.00	1,529,728.75
Productivity (USD)	19,217.22	17,437.34	9,067.28
Stockouts	886.13	724.00	674.99
Value of stocked out inventory (USD)	407,201.41	156,111.70	572,333.94
Inventory value (USD)	479,045.91	464,067.25	372,349.75
Inventory value over sales	0.25	0.23	0.10
Total price changes	14,637.87	12,094.50	13,149.14
Number of suppliers	826.16	869.00	335.13
Number of products	54,875.33	48,178.50	49,284.88

Notes: Table 2 reports means and standard deviations for sales-, pricing-, and inventory-related variables for all stores prior to the arrival of a good manager. All statistics are reported at the store-month level. *Productivity* refers to the sales by worker (sales divided by number of workers). A *stockout* means that a product was sold out in a given month, while *Inventory value* shows the total value of products with positive inventory levels. *Price changes* is measured as the number of times a product has a positive or negative price change between two months. *Low-selling products* are products with sales 1 standard deviation below the mean.

results include all middle managers. Thus, even though our empirical results include all middle management, our survey evidence comes from only the top 30 percent of middle management. While these limitations mean we must take care to not overweight the findings, this survey guided the questions we ask in the empirical analysis, making it a useful guide to the hypotheses we test rigorously below.¹⁰

For example, from the survey we learn that 82% of managers report checking the placement or storage of inventory at least daily, 81% work on worker scheduling at least weekly, 64% work on product restocking at least weekly, and 57% report doing some form of in-store marketing at least weekly. From these statistics it remains unclear whether some managers do more of many different things (or even everything) or whether managers just prioritize differently. While the survey may be able to tell us what all managers do or don't do, on its own, it does little to help us determine what good managers do. Our innovation comes from our ability to combine measures that can be collected only via survey, unusually detailed administrative data on the actions that managers take, and an empirical measure of managerial quality.

In section 4.1, we introduce that empirical measure of managerial quality and discuss

¹⁰The value of this type of analysis is especially clear when compared with the near-zero correlation between the productivity effects of managers and common demographic traits that we (see Appendix Table A.9), and others (Weidmann et al., 2026) observe.

its identification and estimation. We then combine this measure with the survey data in section 4.3 to present descriptive results showing how managers’ self-reported behaviors correlate with the empirical measure of managerial quality. This motivates our causal analysis beginning in section 4.7 which leverages the arrival of good quality managers to stores with poor managerial quality stemming from the quasi-random rotation of managers across stores due to firm policy to identify the specific operational adjustments the arriving manager makes.

4 Empirical Analysis

In the empirical analysis that follows, we first establish the existence of often large but varied manager value-added to sales and productivity using a two-way fixed effect model. We then confirm that the arrival of a high-value-added (good) manager to a store with low-value-added managers (bad store) does, in fact, on average increase store productivity and sales, and document the size and timing of this effect.¹¹ Finally, we uncover the specific operation adjustments managers make to achieve such increases, including how they adapt the organizational structure and the allocation of personnel.

4.1 Measuring Value-Added of a Manager

We start with the well-established observation that individual managers may exert a large influence on productivity (Lazear et al., 2015; Metcalfe et al., 2023) and classify managers according to their value added to store performance in terms of productivity. We estimate this value-added from a two-way fixed effects model of log productivity. Estimates of the managers’ fixed effects would be biased if moves are correlated with time-varying components of store productivity. That is, we are able to leverage the rotation of managers across stores only if the rationale for these rotations is unrelated to manager-store match effects or other transitory shocks. In our context, the rotation of managers across store branches is likely quasi-random (see Section 2), permitting us to estimate the differences in productivity attributable to individual managers (Weidmann et al., 2026). We use the AKM (Abowd et al., 1999) framework to estimate manager and store fixed effects from linear regression of log productivity. Specifically, we estimate the following two-way fixed

¹¹Note that since we defined good managers by their value-added as estimated from the movement of managers across stores, we would expect that when performing a stacked event study around the movement of high value-added managers to stores with low value-added managers at baseline, store performance would, on average, go up. This is not strictly a tautology but we are not surprised to find this result. However, the magnitude and timing of the onset of these effects are of interest and cannot be obtained from the AKM analysis alone. These require event studies as does the investigation of specific changes in operations after the arrival of the new manager.

effects (TWFE) model:

$$\log(\text{productivity})_{ijt} = \theta_i + \psi_{j(i,t)} + \delta_t + \epsilon_{ijt} \quad (1)$$

where $\log(\text{productivity})_{ijt}$, is total log productivity in store j reporting to manager i , in month t ; $\psi_{j(i,t)}$ is a store-fixed effect, δ_t is a month fixed effect, and θ_i is a manager fixed effect.

Following [Card et al. \(2013\)](#), we further decompose ϵ_{ijt} into three components: $\eta_{j(i,t)}$, a match-specific component, ξ_{it} , a unit root component, and ν_{ijt} , a transitory error term:

$$\epsilon_{ijt} = \eta_{j(i,t)} + \xi_{it} + \nu_{ijt} \quad (2)$$

4.2 Identification

Anecdotally, the rotation of managers across stores is mechanical and assignments are made without considering manager traits or store needs. These rotations are coordinated by central headquarters, with the stated rationale being that exposure to different stores is an important part of a manager’s career development.

Formally, valid identification of manager fixed effects requires the assumption that the assignment of managers to stores is (conditionally) mean-independent of past, present, and future values of ϵ_{ijt} . Specifically, managers must not be assigned to stores based either of $\eta_{j(i,t)}$ (a manager-store pair’s match-specific component) or ν_{ijt} (transitory shocks to store performance). Managerial sorting based on either of these terms would mean this sorting is endogenous, resulting in biased and inconsistent estimates. In [Section 4.2.1](#) we check for signs of endogenous sorting via a battery of formal tests, none of which suggest the presence of sorting on the match-specific component.

Notably, our identifying assumption does not restrict the possibility that managers are assigned to stores based on permanent aspects of managerial skill (θ_i) or permanent store features (ψ_j). Thus, even though our partner firm has told us that managers are assigned to stores quasi-randomly, we note that even if sorting on these (permanent) fixed effects does occur (e.g., managers successfully negotiate to work in stores closest to where they live), our identification is still valid.

Finally, these fixed effects are identified only within the set of stores that can be linked by manager(s) moving across stores within the set (i.e., “connected sets”). We observe at least one month of data from at least two different stores for 28 percent of managers. This allows us to identify 6 connected sets. We estimate Equation (1) within each connected set. [Appendix Table A.13](#) shows the results of the AKM decomposition, suggesting that manager fixed effects explain more than 12 percent of log productivity.

4.2.1 Tests for Endogenous Mobility

In addition to assurances from our partner firm, we perform a series of formal tests designed to identify the presence of endogeneity in the movement of managers across stores.

As a first test, we follow [Card et al. \(2013\)](#) to conduct an event study around manager moves between stores. This test is designed to expose a pattern in which managers move due to productivity shocks. We do this by isolating movers and ranking them first by average productivity of the store from which they moved, and second, average productivity of the store to which they moved. We split the average productivity of stores into quartiles. In [Figure A.14](#) we plot the average monthly (residual) productivity of the moving manager on the y-axis. We display only moves away from either the top (quartile 1) or bottom (quartile 4) quartiles.

The intuition behind this test is straightforward. If match-specific components play an important role in determining when and where managers move, we should observe managers systematically relocating precisely when their productivity is unusually high or unusually low—depending on whether the sorting pattern is positive or negative ([Adhvaryu et al., 2024a](#)). Under this scenario, managers who move would tend to gain or lose productivity on average, largely independent of how productive their origin store was. If, instead, moves are not driven by match-specific components, the relationship should look quite different: managers who move to higher-productivity stores should become more productive, while those moving to lower-productivity stores should become less productive ([Adhvaryu et al., 2024a](#)). [Figure A.14](#) confirms this latter pattern. In general, managers moving to better stores gain in terms of productivity, while managers moving to less productive stores become less productive. Moreover, top (bottom) quartile managers moving to top (bottom) quartiles stores experience minimal changes in productivity.

Our next test is also following [Card et al. \(2013\)](#). We plot all possible combinations of origin and destination stores, ranked by quartiles of average manager productivity. [Figure A.15](#) plots the average change in residual log productivity for managers moving from quartile i to quartile j against the corresponding change for managers moving in the opposite direction, from j to i . Under conditional mean independence of the match-specific component, these two changes should be equal in magnitude and opposite in sign—that is, they should lie along the 45-degree line. Any conditional mean dependence arising from the match-specific component would instead appear as systematic deviations from this line. Reassuringly, the deviations we observe in [Figure A.15](#) appear small and non-systematic.

Finally, we consider the distribution of differences between the quality of the arriving manager (as measured by the fixed effect on log productivity that manager exerts in the AKM decomposition) and existing store management (as measured by the mean

of the fixed effects of all existing managers prior to the new manager’s arrival). For example, it seems reasonable to suspect that if our partner firm could observe manager quality they might try to match high-quality managers to help out at struggling stores. Figure A.9 plots the distribution of differences between arriving and existing managers. The distribution is roughly symmetric, suggesting that our partner firm is not matching managers to stores based on manager or store quality.

As a final test, Table 3 (which we present below after formally defining treatment in our context) checks for pre-trends. We find no evidence of systematic differences between treated and untreated stores in the period prior to treatment.

Overall, these tests confirm that managers are not assigned to stores based on their transitory shocks to store performance or match-specific component.

4.2.2 Limited Mobility Bias and Heterogeneity

The correlation between manager and store fixed effects may be subject to limited mobility bias, leading to biased estimates (Abowd et al., 2003; Andrews et al., 2008, 2012). As a first check, we note that 28 percent of managers in our sample move stores at least once. For reference, the share of movers is 12 percent in Andrews et al. (2012), 25 percent in Card et al. (2013), and 14-23 percent in Metcalfe et al. (2023). Next, we present the bias correction of Andrews et al. (2008) in Table A.16. The results are very similar to our baseline model.

We also follow Kline et al. (2020) who use a leave-one-out procedure to develop an unbiased, consistent estimator of variance and covariance terms in two-way fixed effects models, even with heterogeneity in both sets of fixed effects (in our case, at the store and manager levels). We present these results in Table A.16. Again, the results are very similar to our baseline model. These tests suggest limited mobility bias and heterogeneity are unlikely to be issues in our setting.

The results of all these tests suggest that the AKM framework is appropriate in our setting.

4.3 Using manager survey to characterize good managers

We first use the estimates of manager-specific value-added obtained from the AKM estimation in regressions of managerial survey responses. We do this for the subset of relevant questions shown in Figure 1. Each row corresponds to a separate question or index from a set of questions. We run a separate regression for each row, using standardized manager fixed effects (on log productivity) from the AKM decomposition described in Section 4 as the explanatory variable, and survey responses to each question as the

dependent variable. Thus, a large and positive coefficient means there is a strong, positive correlation between the managers' value added to productivity and that survey measure.

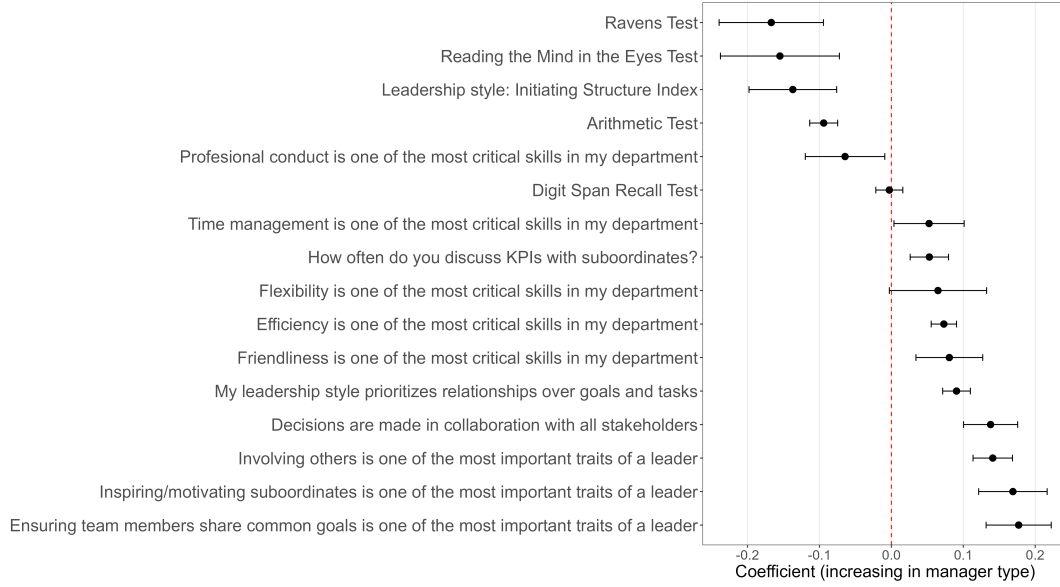
Panel A presents managers' results on standardized tests of cognitive and emotional ability along with self-reported measures of a manager's leadership style. Perhaps surprisingly, we observe that high-value-added managers tend to receive low scores on these tests of cognitive ability and emotional intelligence. On the other hand, higher-value-added managers are more likely to report that traits like friendliness and flexibility are critical skills in their department and that involving, motivating, inspiring, and collaborating with subordinates are important traits in leaders. That is, it seems a high manager value-added is correlated more with a prioritizing of social skills than with the underlying socio-emotional ability of the manager. Panel B focuses on what actions these managers consider most important. Better managers tend to report spending less time on in-store product marketing, not much time on shift scheduling, and more time on inventory management.

Taken together, these results underscore a few themes which appear to differentiate managers who are most likely to increase a store's productivity. First, they do not rank among our partner firm's most intelligent or socially adept (e.g., good managers score lower than average on the Ravens progressive matrices tests, a test of simple arithmetic ability, and even the reading the mind in the eyes test). However, good managers seem to prioritize a strong rapport with their team members and the cohesiveness of the team they manage (e.g., good managers are more likely to list that involving others, ensuring team members share common goals, and inspiring their workers are among the most important traits for leaders; and are more likely to report that their leadership style prioritizes relationships over goals and tasks). Finally, good managers are more likely to prioritize inventory management over product marketing or personnel scheduling (e.g., a good manager is far more likely to say that ordering products is their most important and time-consuming task, and that they check inventory placement daily; but is no more likely to report spending time on worker scheduling, and far less likely to report doing in-store product marketing).

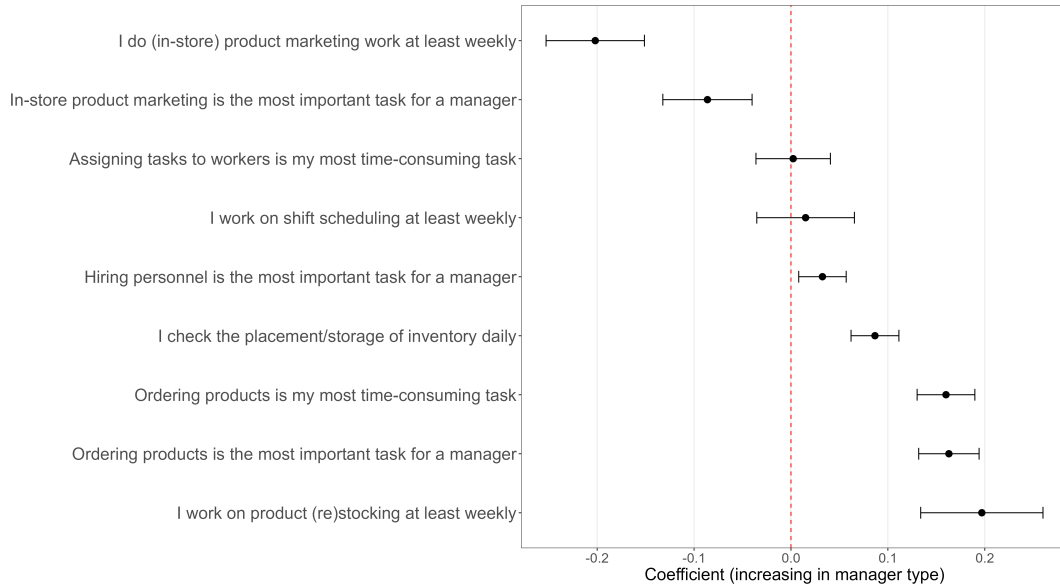
4.4 Definition of Treatment

The above decomposition enables us to classify managers and stores according to their fixed effects on productivity. Specifically, we refer to any manager who exerts an above-median fixed effect (within the connected set) on log productivity as a "good" manager. To classify a store as good or bad, we first take the mean of the fixed effects of all managers who worked in that store immediately prior to the arrival of the new manager. We then compare this mean to the median fixed effect of managers across the 220 stores. Thus,

Figure 1: Survey Results



(a) Ability and Leadership Style



(b) Workplace Priorities

Notes: Each row of Figure 1 corresponds to a question (or index from a set of questions) asked to 183 managers at our partner firm. We run a separate regression for each row, using standardized manager fixed effects (on log productivity) from the AKM decomposition described in Section 4 as the explanatory variable, and survey responses as the dependent variable. Panel (a) presents results relating to managers' self-reported abilities and leadership styles. Panel (b) presents results relating to managers' self-reported priorities in the workplace.

“Bad” stores are those in which the mean fixed effect of all managers in that store is below median across the 220 stores. Following these definitions, there are 1,376 good managers and 145 bad stores.

In the rest of the paper we define treatment as the arrival of a “good” manager to a “bad” store. This is the most intuitive and interesting thought experiment in our context:¹² Bringing a high-performing manager to a low-performing store seems likely to improve performance, while it’s less clear that bringing a good manager to a store that is already performing well should make a difference.¹³

Having defined treatment, we now present the results of a test for the existence of trends prior to the arrival of a good manager to a bad store. Specifically, Table 3 regresses store-level pre-treatment means for each outcome of interest on a dummy for whether a store was ever treated. Systematic differences in the reported coefficients would indicate that treated stores differed from untreated stores in the period prior to treatment. In each case, we observe insignificant results. This supplements our anecdotal evidence and the formal tests for endogenous mobility presented above, suggesting that the rotation of managers across stores is indeed quasi-random.

4.5 Baseline Specification

Our baseline specification is an event-study model (Freyaldenhoven et al., 2021; Roth, 2022). This enables us to test for pre-trends across stores before the arrival of a good manager and measure post-arrival impacts. Treatment is defined, as above, as the first arrival of a high-value-added manager to a low-managerial-quality store. Our preferred specification uses the full store-month panel and retains stores that are never treated during our sample period as the comparison group. We use this as the baseline because never-treated stores provide an untreated benchmark in every calendar month, while the

¹²Weidmann et al. (2026) studies a similar context but uses a different definition of manager moves with the intention of matching the analogous exercise they study in the lab to capture managerial quality.

¹³Even so, we consider the effects of several alternative definitions of treatment in Table A.15. Panel A computes the difference between the fixed effect of the arriving manager and the mean fixed effect of managers already in the store, then sorts these differences in descending order. (See Figure A.9 for the distribution of these differences). Thus, a top 10% difference represents cases in which the arriving manager is much better than the existing management. Indeed, in this case we observe a large and significant increase of 0.185 log points on sales. When we expand treatment to include the top 25% of differences the treatment effect falls to (a still significant) 0.089 log points, and falls further to a (marginally significant) 3.4% increase when we consider the top 50% of differences. Treatment effects are negative when we consider the bottom 50% of differences. Finally, when we consider the arrival of any manager to any store we estimate an insignificant effect. In Panel B we present another alternative definition in which we disregard the quality of existing management in the store and ask whether the arriving manager can increase log sales, regardless of existing store quality. As expected, the effects on log sales are small and insignificant, suggesting that the arrival of a manager who does not differ from the existing managers is unlikely to cause major changes.

Table 3: Check for exogeneity of manager arrival

	Coef.	<i>t</i> -stat	Mean	Obs
Sales (thousands USD)	-196.7730	-0.909	1667.487	220
Productivity (thousands USD)	-0.6415	-0.644	19.490	220
Stockouts	-80.3029	-0.989	871.598	220
Inventory value (millions USD)	-0.0680	-1.389	0.468	220
Inventory value over sales	0.0074	0.519	0.258	220
Price changes	-1960	-1.165	14094.153	220
Products	-7800	-1.158	53013.740	220
Total back-office workers	-1.6059	-0.396	24.577	220
Total customer-facing workers	2.0732	0.173	77.437	220
Total back-office managers	-1.0369	-1.379	5.782	220
Total customer-facing managers	0.3324	0.591	1.920	220
Total hiring (back-office)	-0.1619	-1.831	0.336	220
Total hiring (customer-facing)	0.9240	0.899	4.463	220

Notes: Each row reports the OLS coefficient from a cross-sectional regression of the store-level pre-treatment mean on an indicator for ever being treated (arrival of a good manager to a bad store). We residualize each outcome against month fixed effects to partial out any potential shocks. The regression constant is added back to each residual so the adjusted series is centered at the mean. The full panel of treated and never-treated stores is used. Treated stores contribute only pre-treatment periods to the collapse; never-treated stores contribute all periods. Robust standard errors. Monetary variables expressed in USD.

alternative specification that excludes never-treated stores and identifies effects only from not-yet-treated stores is reported as a robustness check in Section 6.2.

We estimate the following baseline specification:

$$Y_{s,t} = \alpha_0 + \sum_{\underline{C} \leq k \leq \overline{C}, k \neq -1} D_{st}^k \delta_k + \Phi_s + \theta_t + \epsilon_{s,t} \quad (3)$$

where $Y_{s,t}$ is the outcome of interest¹⁴ in store s in month t . For treated stores, D_{st}^k is a relative-time indicator equal to one when store s is k months from the first arrival of a good manager: $D_{st}^k = \mathbb{1}[t = \tau_s + k]$ for $k \in (\underline{C}, \overline{C})$, $D_{st}^{\underline{C}} = \mathbb{1}[t \leq \tau_s + \underline{C}]$, and $D_{st}^{\overline{C}} = \mathbb{1}[t \geq \tau_s + \overline{C}]$, where $\mathbb{1}[\cdot]$ is the indicator function and τ_s is the first monthly period in which the good

¹⁴This estimation strategy introduces potential for “circularity” when studying productivity as an outcome: Since the AKM is a decomposition of log productivity, an event study depending upon the arrival of managers who are classified as good by the AKM will necessarily have a positive effect, provided that the assumption of exogenous mobility is satisfied. Below, when we consider the outcomes of productivity and/or sales we interpret these outcomes as a validation of our AKM decomposition, not as notable results on their own. This problem does not exist for outcomes that do not depend directly upon productivity (e.g., the number of stockouts, the number of trainings conducted, etc.).

manager works at store s . For never-treated stores, τ_s is undefined and all relative-time indicators are set equal to zero in every month. These stores therefore enter the regression through the store fixed effects, Φ_s , and month fixed effects, θ_t , and provide the untreated counterfactual path.

The coefficients δ_k trace the evolution of outcomes in treated stores relative to the month immediately before the good manager arrives and relative to contemporaneous changes among never-treated stores. Coefficients for $k < 0$ provide a test for differential pre-trends, while coefficients for $k \geq 0$ measure the post-arrival effects of the good manager. We normalize $\delta_{-1} = 0$ and set $\underline{C} = -11$ and $\overline{C} = 11$. All specifications include store and month fixed effects, and standard errors are clustered at the store level.

4.5.1 Identification

Identification of the event-study coefficients in Equation (3) relies on three related assumptions. First, stores must not anticipate the arrival of a good manager in ways that change outcomes before the manager arrives. Second, absent the arrival of a good manager, treated stores and never-treated stores would have followed parallel trends after accounting for permanent store differences and common monthly shocks. Third, the timing and destination of manager rotations must not be systematically related to time-varying store shocks or to store-specific needs that would independently affect the outcomes we study.

These assumptions are consistent with the institutional details of the setting. As discussed in Section 4.2, manager rotations are coordinated by central headquarters, are described by the partner firm as mechanical, and are motivated by the firm’s desire to expose managers to different stores as part of career development. The assignment process therefore does not appear to be driven by store-level requests, manager traits, or contemporaneous operational needs.

Our empirical evidence also supports this identifying variation. The endogenous-mobility tests in Section 4.2.1 show no evidence that managers move in response to transitory productivity shocks or match-specific productivity components. In addition, Table 3 compares treated and never-treated stores before treatment and shows no systematic differences in pre-treatment outcomes. We also compare early- and late-treated stores in Table A.14. Finally, the event-study figures reported below provide a direct visual check for differential pre-trends: if treated stores were already improving or deteriorating relative to never-treated stores before the arrival of a good manager, we would expect to see systematic movement in the coefficients for $k < 0$.

Using never-treated stores as the baseline comparison group anchors the counterfactual path in stores that are not exposed to treatment during the sample period, rather than

relying solely on comparisons among stores treated at different dates. The tradeoff is that never-treated stores may differ from treated stores in ways that are not fully captured by store and month fixed effects. For this reason, we report two complementary robustness checks below. First, we estimate event-study specifications that are robust to heterogeneous treatment effects under staggered adoption. Second, in Section 6.2, we exclude never-treated stores and re-estimate the event study using only variation among ever-treated stores, so that identification comes from not-yet-treated stores serving as controls for earlier-treated stores.

4.6 Managerial Quality and Performance

Given the large literature documenting that managerial practices drive performance (e.g., Syverson (2011)), we first confirm that this is true in our context (Weidmann et al., 2026). Table A.10 shows that log sales and log productivity both increase by approximately 12 percent after the arrival of a good manager. The black dots in Panels (a) and (b) of Figure A.10 show that these increases occur shortly after the good manager’s arrival and increase over time. While this is a large increase (equivalent to nearly 100 times the monthly salary of the average manager), we do not consider it surprising since the event study framework is constructed using the manager fixed effects derived from the AKM decomposition. Thus, we interpret these findings as validation that the AKM decomposition we conduct is properly identifying these managers’ effect on productivity. Having verified that we are properly identifying good managers we next show how managers unlock higher productivity.

4.7 What do good managers do?

We find that good managers improve store performance through three operational levers. First, they improve inventory management—*reducing stockouts while making inventory leaner*. Second, they *make better (revenue-increasing) price changes* without changing how often prices move. Third, they *rebalance labor toward back-office roles* related to inventory, pricing, and supply, with no increase in overall headcount. We document each in turn.¹⁵

4.7.1 Action 1: Good managers reduce stockouts and make inventory leaner

Prior to the arrival of a good manager, bad stores experience an average of 763 stockouts per month. Good managers cause their stores to experience 88 (or 11.5 percent of the

¹⁵We deal with revenue instead of profit due to the data limitations described in Section 3. Even so, in the retail context revenue is an excellent proxy for profits so long as most goods are sold for a weakly positive profit.

regression-sample mean) fewer stockouts per month (see column 1 of Table 4). Figure A.11 shows that the number of stockouts begins to drop shortly after the arrival of a good manager and continues to fall over time.

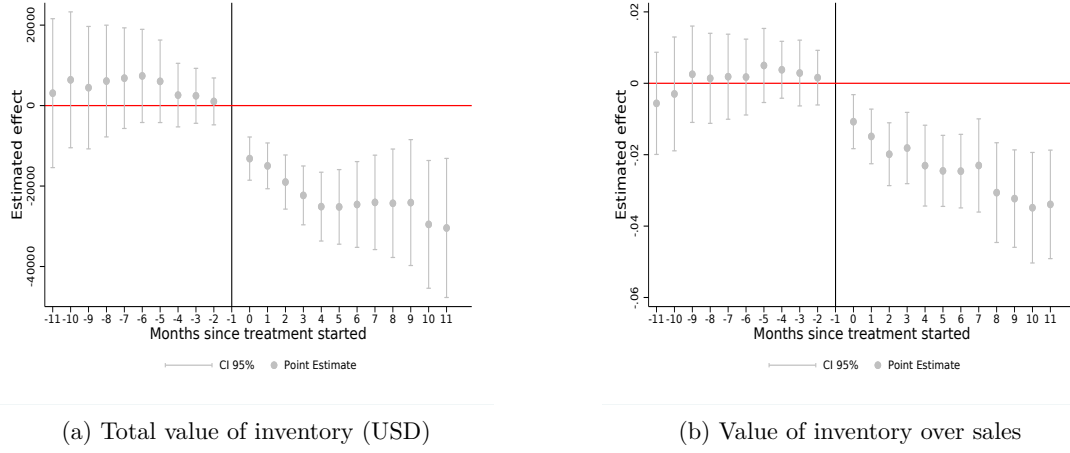
Table 4: Effects on stockouts and on-hand inventory value

	(1) Stockouts	(2) Inventory Value (USD)	(3) Inventory Value over Sales
Treatment effect	-87.82*** (32.59)	-28,996*** (8,779)	-0.0290*** (0.00579)
Observations	8,890	8,890	8,890
R-squared	0.886	0.986	0.808
Treated Stores	87	87	87
Stores	220	220	220
Mean of Dv	763.3	404,962	0.247
Relative Effect	-11.5%	-7.2%	-11.7%

Notes: Table 4 shows the point estimates of stockouts and on-hand inventory. A stockout is defined as a product whose number of items is equal to or below zero. The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. The treatment effect is defined as the arrival of a good manager to a bad store (i.e., a store in which the average fixed effect of incumbent managers is below that of the median store). Robust standard errors are in parentheses. * significant 10%, ** significant 5%, *** significant 1%.

Column 2 of Table 4 shows that the arrival of a good manager causes a drop of approximately \$29K USD (a 7 percent drop) in the overall value of a store’s inventory. Panel (a) of Figure 2 shows that this drop occurs in the period after the manager’s arrival and that inventory values further decrease post-arrival. Column 3 of Table 4 shows that the inventory value relative to sales drops by nearly 12 percent. This drop is especially impressive when considering the previously documented increase in sales and decrease in stockouts, both of which seem likely to cut against the existence of a smaller inventory. Panel (b) of Figure 2 shows that this pattern too intensifies over time: the longer a good manager stays at a bad store, the lower the ratio of inventory value to sales.

Figure 2: Value of inventory



Notes: This figure shows event-study coefficients and 95 percent confidence intervals from estimating the effect of a good manager's arrival on (a) the total value of inventory (USD) and (b) the value of inventory divided by total sales. The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. Event time is measured in months relative to the arrival of a good manager, with the month immediately prior to arrival normalized to zero. All specifications include store and month fixed effects, and standard errors are clustered at the store level. The vertical line indicates the month of a good manager's arrival.

4.7.2 Action 2: Good managers make better pricing decisions

Table 5 presents the effects of the arrival of a good manager to a bad store on pricing-related outcomes. Column 1 shows that overall log sales rise by 11.7 percent. Column 2 shows that, among products that experience a price change, log sales rise by approximately 17 percent. Columns 3 and 4 show that this effect comes from a combination of revenue gains following price increases (column 3) and price decreases (column 4); the large, significant increase even when prices are raised (column 3) helps rule out the possibility that managers are simply achieving higher sales by lowering prices. Column 5 shows that sales also rise for products without a price change, by approximately 6.2 percent. Sales gains among products with price changes are thus nearly three times as large as among products without. Column 6 shows that good managers do not make *more* price changes than were made before their arrival—they make *better* price changes. Figure A.12 presents the event-study analogs.

4.7.3 Action 3: Good managers rebalance labor across departments

Prior to arrival of a good manager, bad stores employ 24 employees in back-office departments, roughly 6 of whom are managers, whereas customer-facing departments employ more than three times that number, but with less than half as many managers.

Table 5: Pricing decisions

	(1) Log sales	(2) Log sale products with price change	(3) Log sale products with positive price change	(4) Log sale products with negative price change	(5) Log sale products without price change	(6) Num price changes
Treatment effect	0.117*** (0.0218)	0.169*** (0.0310)	0.161*** (0.0369)	0.144*** (0.0287)	0.0617** (0.0284)	-3.618 (247.6)
Observations	9,007	8,748	8,541	8,748	8,749	9,007
R-squared	0.984	0.918	0.855	0.845	0.878	0.930
Treated Stores	87	87	87	87	87	87
Stores	220	220	220	220	220	220
Mean of Dv	21.73	18.04	17.29	17.28	15.91	13,108

Notes: Table 5 shows point estimates on the log sales of any product (column 1), the subset of products that had any price change (column 2), a price increase (column 3), a price cut (column 4), or no price change (column 5). Column (6) shows the effect of treatment on the number of price changes. The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. The treatment effect is defined as the arrival of a good manager to a bad store (i.e., a store in which the average fixed effect of incumbent managers is below that of the median store). Robust standard errors are in parentheses. * significant 10%, ** significant 5%, *** significant 1%.

Column 1 of Table 6 shows that the arrival of a good manager to a bad store causes back-office departments to gain an average of 1.3 employees. The event study analog to this result (Panel (a) of Figure 3) shows that the growth in these back-office departments begins shortly after the arrival of a good manager and increases in all periods thereafter. More than 1 in 4 of these new employees are managers, with the average back-office department gaining 0.36 new managers. In customer-facing departments, neither the total workforce nor the number of managers changes significantly.

In Table A.12 (and the analogous event study figures in Figure A.13) we probe the mechanisms underlying these organizational changes. The increase in back-office managers comes from significant reductions in turnover (back-office departments retain 0.12 more managers) and increases in hiring (back-office departments hire 0.19 more managers). Very little movement of managers to back-office departments occurs. For customer-facing departments we observe no effects on turnover, hiring, or moves of managers.

Table 6: Workforce in back-office and customer-facing departments

	(1) Back-office workforce	(2) Customer-facing workforce	(3) Back-office managers	(4) Customer-facing managers
Treatment effect	1.310** (0.543)	-0.423 (1.494)	0.358** (0.146)	0.175 (0.112)
Observations	9,007	9,007	9,007	9,007
R-squared	0.990	0.980	0.972	0.964
Treated Stores	87	87	87	87
Stores	220	220	220	220
Mean of Dv	24.81	77.84	5.496	2.387
Relative Effect	5.3%	-0.5%	6.5%	7.3%

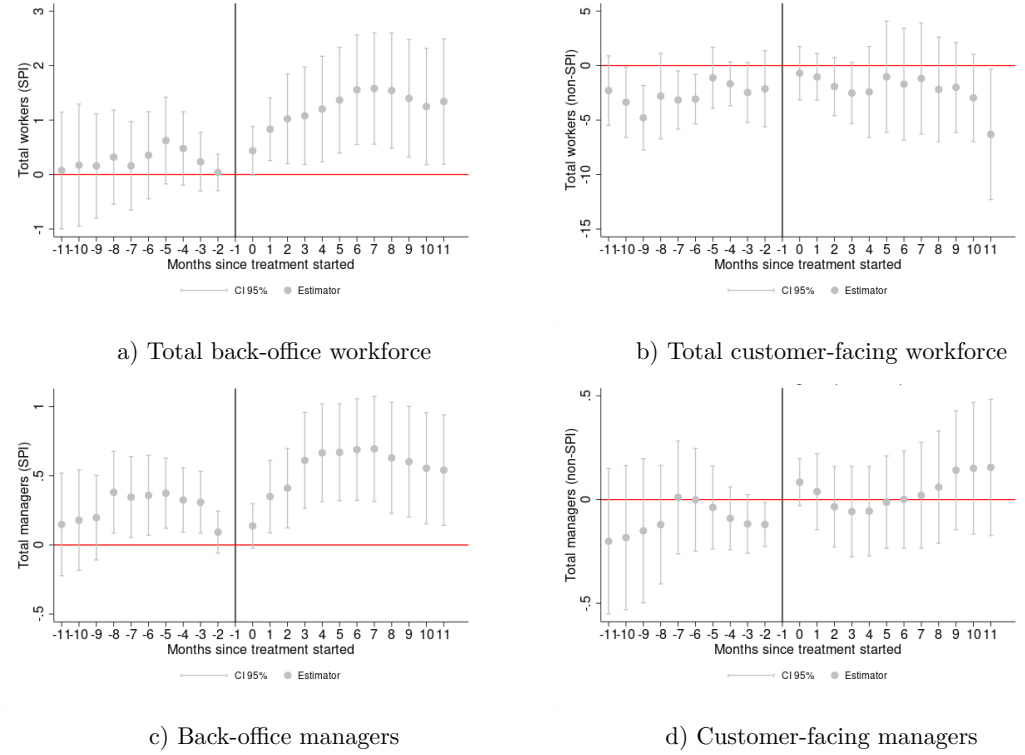
Notes: Table 6 shows the point estimates of total workforce, number of managers, and the ratio of managers to non-managers in back-office and customer-facing departments. Back-office refers to departments whose main responsibilities are related to sales, pricing or inventory activities, while customer-facing departments are those with main responsibilities other than those mentioned above. The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. The treatment effect is defined as the arrival of a good manager to a bad store (i.e., a store in which the average fixed effect of incumbent managers is below that of the median store). Robust standard errors are in parentheses. * significant 10%, ** significant 5%, *** significant 1%.

4.8 How persistent are these changes?

In this section we study which of the changes implemented by good managers persist after they leave, and for how long. We modify our baseline specification (Equation (3)) to capture the departure rather than the arrival of a good manager. We focus on departures from bad stores only, for the same reasons we focus on arrivals to bad stores: these are the stores with the most scope both for improvement at arrival and for reversion after departure. This is also largely the same set of stores we study in our primary analysis: even after the 12 percent productivity increase that follows the arrival of a good manager, 82 percent of stores that were bad before treatment remain bad after treatment. As before, we control for arrivals of other managers, and additionally for departures of other managers.

The blue squares in Figure 4 trace the effects of a good manager’s departure on log sales (Panel (a)) and log productivity (Panel (b)) at a bad store; black dots reproduce the arrival effects for comparison. Performance begins slipping shortly after departure and stabilizes about four months later. The departure treatment effect on log sales is approximately -6.8 percent, eroding roughly 65 percent of the arrival gain; the departure

Figure 3: Workforce in back-office and customer-facing departments



Notes: This figure presents event-study coefficients and 95 percent confidence intervals for outcomes related to the total workforce and number of managers in back-office and customer-facing departments following the arrival of a good manager. Panels (a) and (b) show the total workforce in back-office and customer-facing departments, respectively. Panels (c) and (d) show the number of managers in back-office and customer-facing departments, respectively. The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. Event time is measured in months relative to the arrival of a good manager, with the month immediately prior to arrival normalized to zero. All specifications include store and month fixed effects, exclude the moving manager, and cluster standard errors at the store level. The vertical line indicates the month of a good manager's arrival.

effect on log productivity is similar in magnitude. Throughout this section we report the *share eroded* for each outcome, defined for logged outcomes as

$$\text{Share Eroded} = 1 - \frac{(1 + a)(1 + d) - 1}{a},$$

where a is the arrival treatment effect and d the departure effect, and as $-d/a$ for outcomes in levels. A value of zero means the arrival effect is fully preserved; a value of one means it is fully reversed; negative values mean the arrival effect grows after departure. Panel (b) of Table 7 reports these shares for each pricing outcome.

We now turn to each of the three actions identified in Section 4.7 and ask whether its associated gain persists.

Stockouts persist. The arrival of a good manager reduced stockouts by 11.5 percent (section 4.7.1). The departure of a good manager has no statistically significant effect on stockouts (Panel (a) of Figure 5); the lower stockout level achieved during the manager’s tenure remains in place after they leave. We interpret this as evidence that the operational improvements driving lower stockouts—better inventory monitoring, supplier coordination, and related practices—are durable, plausibly because they are embedded in the back-office staffing reallocation we document in Section 4.7.3 (which itself persists; see below).

Inventory leanness persists. Panel (b) of Figure 5 shows that the post-departure monthly point estimates on the value of inventory over sales are positive but insignificant, suggesting that the leaner inventory levels achieved under a good manager are mostly preserved after departure.

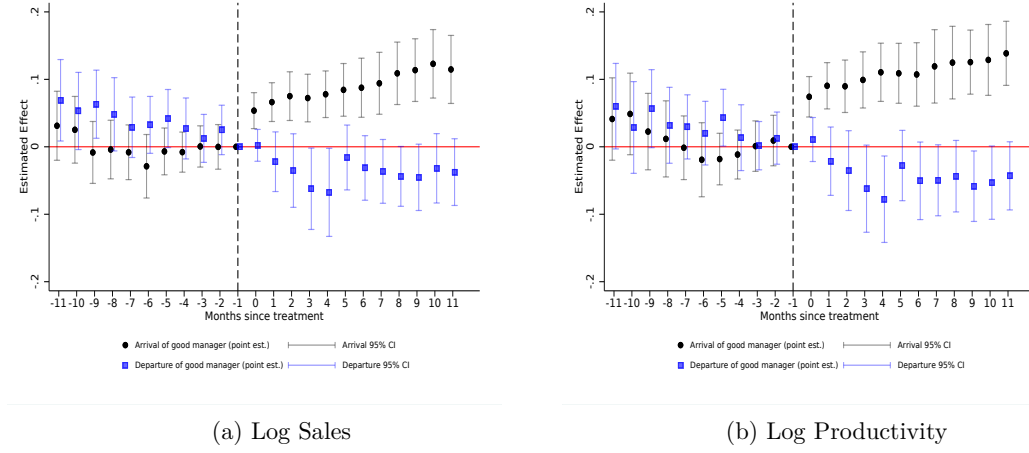
Pricing gains erode. Table 7 reports event-study estimates of departure on the same pricing outcomes shown in Table 5; Panel (b) reports the share of each arrival effect that erodes. Overall log sales fall by 6.8 percent after departure (column 1), eroding roughly 65 percent of the arrival gain. The departure effect on the log sales of products with a price change (column 2) is negative but not statistically significant; the share of the arrival effect that erodes is roughly 40 percent. Patterns are similar for price increases and price decreases separately (columns 3 and 4). By contrast, sales of products without a price change (column 5) do not change significantly after departure; the arrival effect on this margin was only marginally significant in the first place, but what little there was is preserved. The number of price changes (column 6) does not change after departure, just as it did not change at arrival. The reversal of overall sales gains is thus concentrated among products subject to price changes, while sales of products not subject to price changes are largely preserved.

Back-office staffing persists. Unlike at arrival, departure of a good manager is not associated with significant restructuring of the store. The rebalancing of personnel toward back-office roles documented in section 4.7.3 remains in place after the good manager leaves.

Together, these results paint a picture of differential persistence. The components of managerial value-added that are tied to organizational structure—back-office staffing and the lower stockout levels enabled by it—are durable. The component tied to ongoing context-specific judgment—identifying which products warrant a price adjustment in response to evolving local demand—depends on the manager’s continued presence and erodes when the manager leaves. We return to the implications of this distinction in

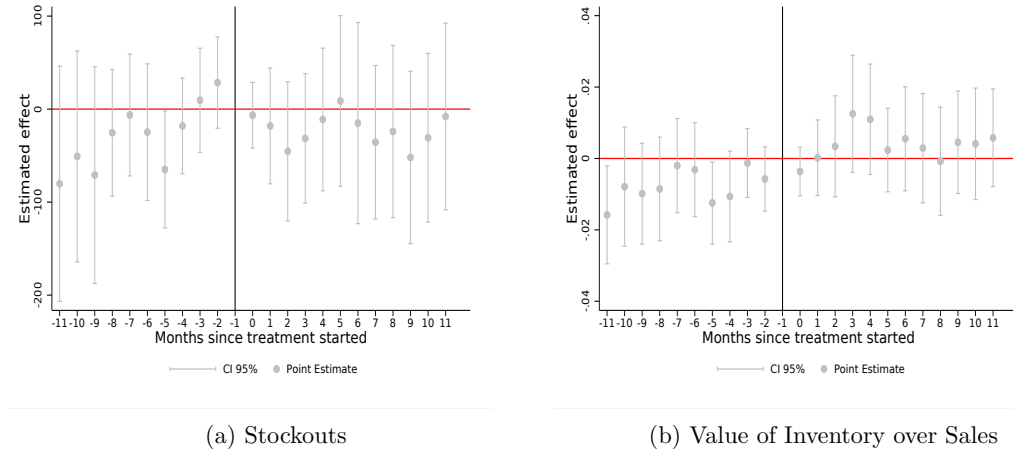
section 7.

Figure 4: Effects of Manager Departure on Store Performance



Notes: The black dots and lines correspond to the event-study coefficients and 95 percent confidence intervals from estimating a good manager's *arrival* to a bad store on log sales and log productivity. The blue squares and lines are the analogous results when estimating a good manager's *departure* from a bad store. The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. The vertical line represents the arrival or departure of a good manager from a bad store.

Figure 5: Effects of Manager Departure on Stockouts and Value of Inventory on Hand



Notes: This figure shows the event-study coefficients and 95 percent confidence intervals from estimating a good manager's *departure* on the number of stockouts and the value of inventory in a store (in USD). The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. The vertical line represents the departure of a good manager.

Table 7: Effects of manager departure on pricing

	(1) Log sales	(2) Log sale products with price change	(3) Log sale products with positive price change	(4) Log sale products with negative price change	(5) Log sale products without price change	(6) Num price changes
<i>Panel (a): Manager Departures</i>						
Treatment effect	-0.0676*** (0.0253)	-0.0570 (0.0370)	-0.0621* (0.0376)	-0.0483 (0.0426)	0.0241 (0.0307)	81.84 (304.5)
Observations	9,007	8,748	8,541	8,748	8,749	9,007
R-squared	0.984	0.917	0.854	0.845	0.878	0.930
Mean of Dv	21.86	18.21	17.46	17.43	15.95	12,582
<i>Panel (b): Share of arrival effect that erodes after departure</i>						
Share Eroded	0.64	0.39	0.45	0.38	-0.41	N/A

Notes: Panel (a) of Table 7 shows point estimates on the log sales of any product (column 1), the subset of products that had any price change (column 2), a price increase (column 3), a price cut (column 4), or no price change (column 5). Column (6) shows the effect of treatment on the number of price changes. The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. Treatment is defined as the *departure* of a good manager from a bad store (i.e., a store in which the average fixed effect of incumbent managers is below that of the median store). Panel (b) presents the share of the *arrival* treatment effect (these values are reported in Table 5) that dissipates (erodes) when a manager departs. The share of the treatment effect that erodes for outcomes reported in levels (e.g., column 6) is simply $-d/a$, where d corresponds to the departure treatment effect, and a corresponds to the arrival treatment effect. For logged outcomes, the simple ratio of coefficients is a poor approximation, so we instead compute $Share\ Eroded = 1 - \frac{(1+a)(1+d)-1}{a}$. A value of 0 means that none of the arrival effects erode, a positive value means some erosion occurs, and a negative value means the arrival effects continued to grow after the manager departed. We suppress results in which the denominator (i.e., the arrival effect, or a) is not significantly different from 0 at the 95% level. Robust standard errors are in parentheses. * significant 10%, ** significant 5%, *** significant 1%.

5 Alternative Stories

Our preferred interpretation of the results presented is that the increased sales and productivity are the direct result of the arrival of a manager who manages his/her resources (personnel, attention, etc.) more effectively. We have shown that the arrival event triggers three operational changes: (1) better inventory management (stockouts fall even as inventory becomes leaner), (2) more effective pricing (the same frequency of price

changes generates higher sales), and (3) a reallocation of labor toward the back-office functions responsible for inventory, pricing, and supplier management.

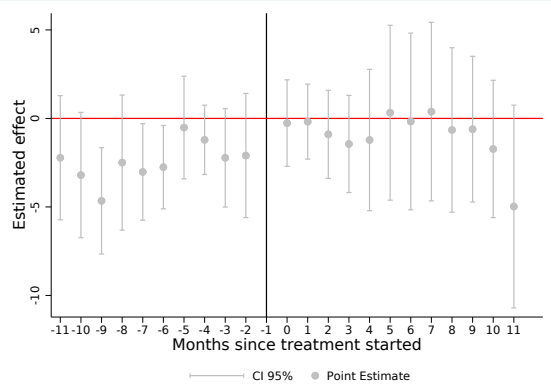
While the most obvious explanation is that all of these changes are the direct result of manager actions, in this section we address several mechanisms other than our preferred interpretation that might explain the observed results. We provide additional evidence supporting our preferred interpretation.

5.1 Strategic Expansion of Stores

The most intuitive alternative explanation of the large increase in sales following the arrival of a good manager to a bad store is simply that the stores in which sales increased were stores that our partner firm had strategically chosen to expand. In fact, we find no evidence of strategic expansion at treated stores, and even find some evidence of reductions in store inventory. Specifically, the following findings lead us to reject the possibility that our results might be explained by a pre-planned expansion of stores:

First, Figure 6 shows that the total number of workers post-treatment does not increase. Second, Figure 2 shows that the arrival of a good manager to a bad store actually causes both the total value of inventory (panel (a)) and the ratio of inventory to sales (panel (b)) to steadily decrease over time.

Figure 6: Number of Workers



Notes: This figure shows event-study coefficients and 95 percent confidence intervals from estimating the effect of a good manager's arrival on the total number of workers in a store. The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. Event time is measured in months relative to the arrival of a good manager, with the month immediately prior to arrival normalized to zero. All specifications include store and month fixed effects, and standard errors are clustered at the store level. The vertical line indicates the month of a good manager's arrival.

5.2 Demand for Additional Manager(s)

Another possible explanation is that the effects we see following the arrival of a good manager are driven by problems preceding the arrival. If, for example, a specific store were struggling to manage inventory they might request (or central management could even proactively assign) an additional manager. While our conversations with the partner firm lead us to believe that that happens only rarely, we address the possibility as if it were a common occurrence.

In order for this to explain the results, we should see a dip (or rise) in the outcome of interest prior to treatment (Ashenfelter, 1978). Throughout the paper we present event study plots for every outcome of interest. Reassuringly, we see no evidence of pre-trends that might be consistent with store-level demand for a manager. As further confirmation that the results are not caused by pre-treatment problems, we also test for the existence of pre-trends (see Table 3).

5.3 Are these results driven by arrival effects?

Another possible explanation for the results we observe is that the large productivity gains are not a result of improved management, but simply the result of new management. In other words, it seems plausible that the arrival of a new manager might unlock an increase in productivity if either that manager works harder to make a good impression and/or workers work harder to please their new manager.

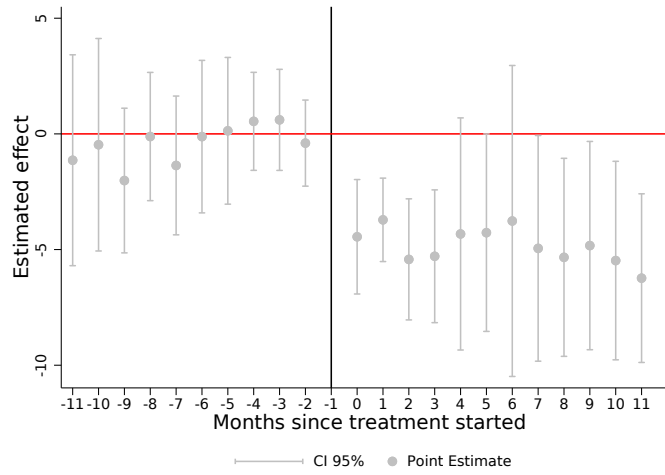
However, if this were happening in our context we would expect to see an increase in sales even when the arriving manager does not improve the management of the store (i.e., when a bad manager arrives to a good store). Table A.15 shows that this is not the case. In fact, when the difference between the arriving manager’s and the mean of the existing managers’ fixed effects is below-median (column 4) we see a significant *decrease* in sales. When we look at the effect of the arrival of any manager to any store we estimate an insignificant, negative effect (column 5).

5.4 Do good managers just do more of everything?

In this section we ask whether good managers are simply better at everything or whether they sacrifice attention to other tasks in order to achieve the better pricing and inventory outcomes we document above. Our survey results led us to hypothesize that good managers may be dedicating less attention to personnel management. Consistent with this, we see that good managers reduce the number of trainings a store conducts by approximately 7 trainings per month (point estimate -7.18 , significant at the 5% level). Figure 7 shows that this reduction begins immediately following the arrival of a new

manager. Because we do not directly observe scheduling, absenteeism is the closest proxy for how much attention a manager allocates to worker scheduling. Similarly, turnover is the best proxy in our dataset for how unhappy workers are generally. Appendix Table A.11 shows that absenteeism rises (by 16 percent, significant at the 1% level), while turnover remains unchanged following the arrival of a good manager. We interpret these findings as suggestive evidence that even if good managers do reduce attention to personnel matters in order to pay attention to inventory and pricing matters, they still pay enough attention to these personnel matters to not directly affect the size or composition of their store’s workforce via attrition.

Figure 7: Effect on number of trainings



Notes: This figure shows event-study coefficients and 95 percent confidence intervals from estimating the effect of a good manager’s arrival on the number of trainings conducted in a store each month. The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. Event time is measured in months relative to the arrival of a good manager, with the month immediately prior to arrival normalized to zero. All specifications include store and month fixed effects, and standard errors are clustered at the store level. The vertical line indicates the month of a good manager’s arrival.

6 Robustness

In this section, we perform several alternative estimations to test the robustness of our baseline event-study specification.

6.1 Heterogeneous Effects

Because our treatment timing is staggered, we must account for the possibility that conventional two-way fixed-effects difference-in-differences estimators place negative or otherwise non-convex weights on underlying cohort-specific treatment effects (Athey and Imbens, 2021; De Chaisemartin and d’Haultfoeuille, 2020; Goodman-Bacon, 2021). If treatment effects vary across stores over time or across cohorts treated in different months, a pooled event-study specification may contaminate the estimated leads and lags by implicitly comparing already-treated stores to newly treated stores (Sun and Abraham, 2020). This concern is particularly relevant in our setting because the arrival of a good manager occurs at different points in calendar time across treated stores.

We address these issues by estimating two alternative event-study specifications that are robust to heterogeneous treatment effects under staggered adoption, using never-treated stores as the comparison group. This matches our baseline specification, in which stores that never receive a good manager during the sample period provide the untreated counterfactual path. First, we estimate the cohort-specific event-study estimator proposed by Sun and Abraham (2020). This approach allows the dynamic treatment effects to vary across treatment cohorts and then aggregates cohort-specific effects into relative event time. In this specification, never-treated stores serve as the reference cohort. Second, we estimate group-time average treatment effects following Callaway and Sant’Anna (2020), where each group is defined by the month in which a store first receives a good manager. We then aggregate these group-time average treatment effects into event time, again using never-treated stores as the control group.

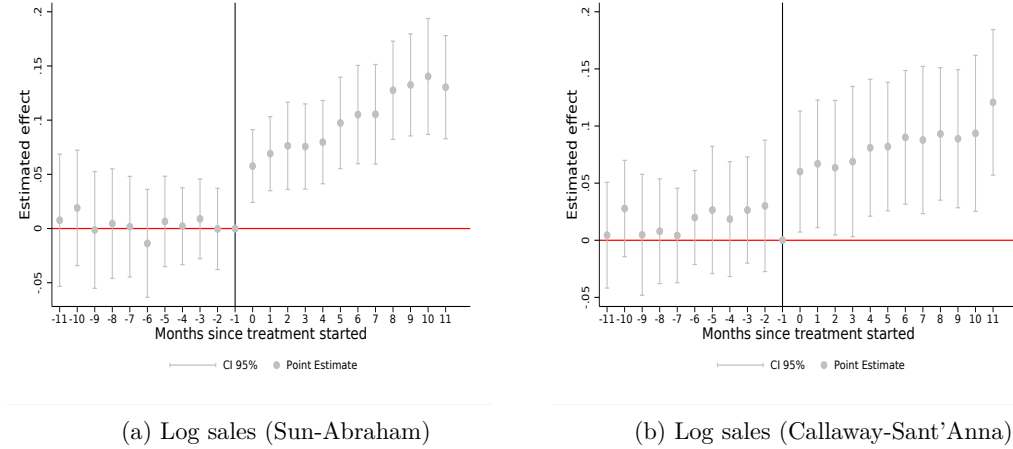
The identifying assumption for both exercises is that, absent the arrival of a good manager, treated stores would have followed the same outcome trends as never-treated stores after accounting for time-invariant store differences and common monthly shocks. This is the same identifying comparison that underlies our baseline event-study specification. Section 6.2 reports a complementary robustness check that excludes never-treated stores and instead uses not-yet-treated stores as the comparison group.

The results of these robustness checks for log sales are presented in Figure 8 (see Appendix Figure A.16 for log productivity). Panel (a) presents the Sun-Abraham specification and Panel (b) presents the Callaway-Sant’Anna specification. These event studies are very similar to the baseline event study reported in Figure A.10, arrivals panel. In particular, the estimated post-arrival effects remain positive and economically meaningful, while the pre-treatment coefficients show no evidence of differential pre-trends.

For both alternative specifications, the point estimates are similar to those from our baseline event-study specification. The treatment effect on log sales is 0.117 in the baseline specification, compared with 0.101 using the Sun and Abraham (2020) specification

and 0.104 using the [Callaway and Sant’Anna \(2020\)](#) specification. For log productivity, the baseline treatment effect is 0.112, compared with 0.113 using the Sun-Abraham specification and 0.110 using the Callaway-Sant’Anna specification.

Figure 8: Heterogeneity-robust Event Studies: log sales



Notes: This figure shows the event-study coefficients and 95 percent confidence intervals from estimating the effect of a good manager’s arrival on log sales using estimators that are robust to heterogeneous treatment effects under staggered adoption. Panel (a) reports the cohort-specific event-study estimates following [Sun and Abraham \(2020\)](#). Panel (b) reports group-time average treatment effects following [Callaway and Sant’Anna \(2020\)](#), translated into relative event time. The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. Event time is measured in months relative to the arrival of a good manager, with the month immediately prior to arrival normalized to zero. The vertical line represents the arrival of the good manager.

6.2 Not-yet-treated Stores

As a robustness check, we re-estimate the baseline event-study specification after excluding the never-treated stores. This specification restricts the sample to stores that receive a good manager at some point between January 2017 and June 2020. Identification therefore comes from variation in the timing of good-manager arrivals across ever-treated stores: stores that have not yet received a good manager serve as controls for stores that receive one earlier.

This exercise addresses the concern that never-treated stores may differ systematically from treated stores along dimensions that are difficult to fully account for with observables and fixed effects alone. Restricting the sample to ever-treated stores aligns the comparison group with stores that are plausibly on a similar managerial rotation path and that are all eventually exposed to the same treatment margin. The cost is that this design relies on a different parallel-trends assumption: in the absence of treatment, earlier- and later-treated

stores would have evolved similarly after accounting for store and month fixed effects. The specification is therefore complementary to our baseline: the baseline anchors the counterfactual path using stores that are never treated, while this robustness exercise asks whether the same patterns emerge when the comparison group is restricted to stores that are eventually treated.

Figure A.17 presents the event-study estimates for sales and productivity using not-yet-treated stores as the control group. The results closely mirror the baseline estimates. Sales and productivity rise after the arrival of a good manager, while the pre-treatment coefficients show no evidence of systematic differential trends. Table A.17 shows that the inventory-related results are also robust to this alternative control group. The arrival of a good manager reduces stockouts by 61.9 per month, or 8.2 percent of the dependent-variable mean. The estimate is smaller and less precise than in the baseline specification, but it has the same sign and remains economically meaningful. The effects on inventory are especially stable across specifications: inventory value falls by approximately \$28,500, or 7.1 percent of the mean, and the ratio of inventory value to sales falls by 0.016, or 6.3 percent of the mean. Figures A.18 and A.19 show that the dynamic patterns are also similar to the baseline results. Inventory becomes leaner after the arrival of a good manager, and the improvements in stockouts and inventory management do not sharply reverse after the manager departs.

The organizational results are likewise qualitatively unchanged. Figure A.20 shows no evidence that the arrival of a good manager leads to a broad expansion of total staffing, consistent with the baseline finding that the performance gains are not driven by simple scale expansion. Figure A.21 shows that the reduction in trainings remains visible in the not-yet-treated specification, consistent with the interpretation that good managers reallocate attention away from some personnel-management activities and toward inventory, pricing, and back-office operations. Finally, Figure A.22 shows that the changes in back-office turnover and hiring remain similar in sign and timing to the baseline results, while the corresponding customer-facing outcomes remain comparatively muted.

Overall, these results show that the main findings are not sensitive to the choice of comparison group. The same broad pattern emerges when identification comes only from timing variation among ever-treated stores: good managers raise sales and productivity, reduce stockouts, make inventory leaner, and reorganize activity toward back-office functions. The similarity between the never-treated and not-yet-treated specifications strengthens the interpretation that the results reflect the arrival of high-value-added managers rather than differential trends between treated and never-treated stores.

7 Discussion

We study what good managers do and whether their impacts persist, exploiting a manager rotation policy to estimate manager value-added and bringing unusually rich administrative data to bear on the operational decisions that drive it. Good managers reduce stockouts while making inventory leaner, generate larger revenue gains from price changes without increasing repricing frequency, and shift labor toward back-office functions related to inventory, pricing, and supply management. When they leave, these gains persist differentially. Staffing rebalances and inventory improvements remain in place; pricing-related gains erode, with the overall sales reversal concentrated entirely among repriced goods. These patterns reveal a fundamental distinction: some managerial contributions are embedded in organizational structure and outlast the manager, while others depend on the manager’s ongoing, context-specific judgment and erode when that judgment is withdrawn.

Our survey evidence complements these findings. Good managers report spending more time on inventory-related procedures and are more likely to identify pricing as a critical determinant of store performance, consistent with the operational margins through which their value-added operates.

These findings raise important questions for future research. One concerns how the codifiability of managerial value-added depends on the production environment. Settings characterized by repetition, stability, and narrower product scope may be more conducive to embedding managerial improvements in durable routines, whereas settings like retail and food service, where demand is variable and the product space large and continuously changing, may leave a larger role for continued managerial adaptation. A second concerns the extent to which AI-based decision-support tools will substitute for the more codifiable components of good management, help codify what currently appears difficult to institutionalize, or instead free managers to specialize in the context-specific judgment that proves hardest to replicate. Clarifying these boundaries remains an important task for future work.

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A Appendix

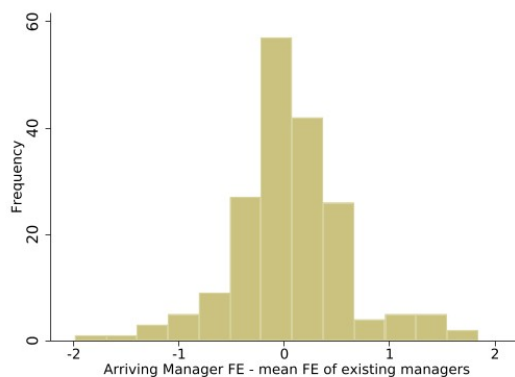
A.1 Additional Statistics

Table A.8: Distribution of Managers

Type	Frequency	Percent
Director	228	8.25
Boss	300	10.85
Leader	1,412	51.07
Coordinator	825	29.84
Total	2,765	100

Notes: Table A.8 shows the distribution of each type of manager in descending order of seniority. Directors are the most senior manager type while coordinators are the most junior.

Figure A.9: Distribution of differences between arriving and existing management



Notes: This figure shows the distribution of the differences between the fixed effect of the arriving manager and the mean of the fixed effects of all managers working at the store prior to the arrival of the new manager.

Table A.9: Correlation with Manager Fixed Effect

	Correlation
Female	0.0253
Age	-0.0261
Tenure (weeks)	0.0086
Employee Level	0.0023
Married	-0.0103
Single	0.0044
Civil Union	0.0036

Notes: Table A.9 reports correlations between the fixed effect of managers and the demographic traits of all managers for whom we estimate the fixed effects.

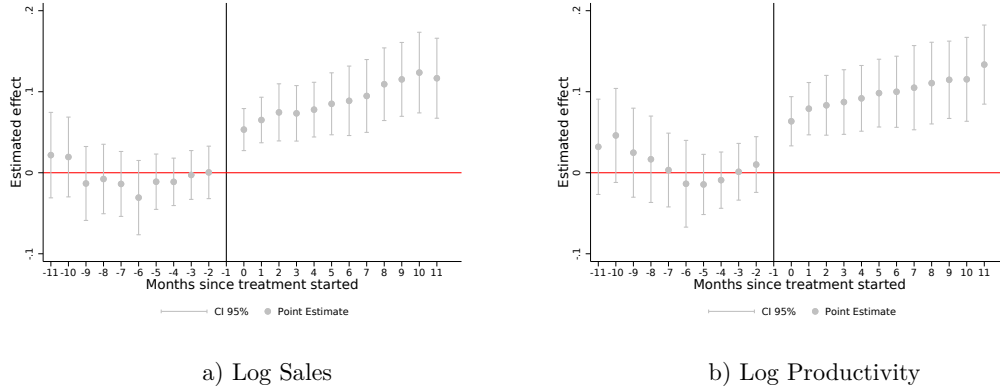
A.2 Additional Evidence

Table A.10: Treatment effects on log sales and log productivity

	(1) Log sales	(2) Log productivity
Treatment effect	0.117*** (0.0218)	0.112*** (0.0217)
Observations	9,007	9,007
R-squared	0.984	0.818
Treated Stores	87	87
Stores	220	220
Mean of Dv	21.73	17.91

Notes: Table A.10 shows the point estimates of log sales and log of productivity (measured as sales over number of workers per store). The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. The treatment effect is defined as the arrival of a good manager to a bad store (i.e., a store in which the average fixed effect of incumbent managers is below that of the median store). Robust standard errors are in parentheses. * significant 10%, ** significant 5%, *** significant 1%.

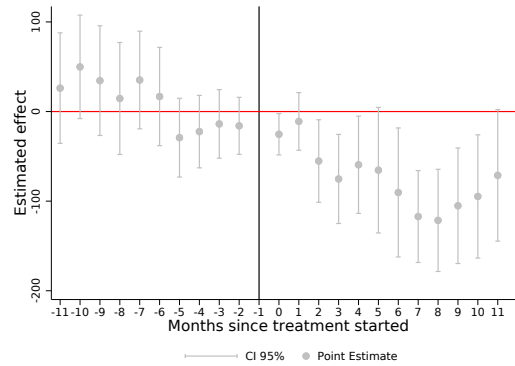
Figure A.10: Effect of a Good Manager on Sales and Productivity



Notes: Each panel of this figure shows the event-study coefficients and 95 percent confidence intervals from estimating a good manager's arrival on Sales and Productivity. The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. The vertical line represents the arrival of a good manager.

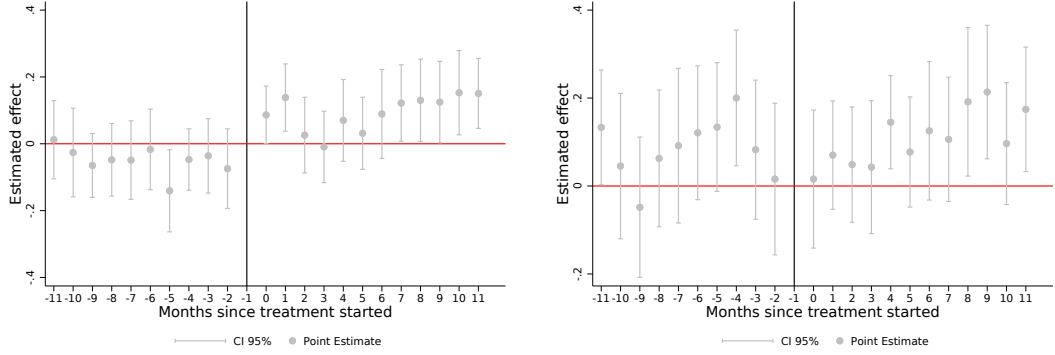
A.2.1 Additional Evidence on Manager Actions

Figure A.11: Effects on the Number of Stockouts



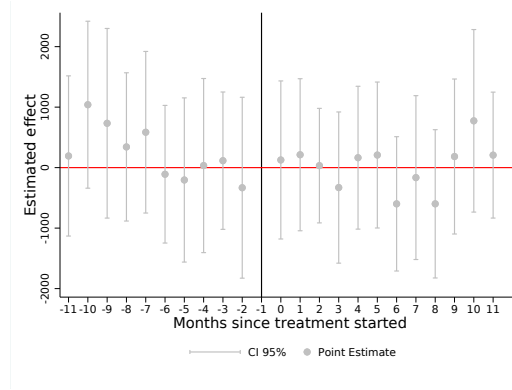
Notes: This figure shows event-study coefficients and 95 percent confidence intervals from estimating the effect of a good manager's arrival on the number of stockouts. The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. Event time is measured in months relative to the arrival of a good manager, with the month immediately prior to arrival normalized to zero. All specifications include store and month fixed effects, and standard errors are clustered at the store level. The vertical line indicates the month of a good manager's arrival.

Figure A.12: Price Changes



(a) Log sales of products *with* a price change

(b) Log sales of products *without* a price change



(c) Number of price changes

Notes: This figure shows event-study coefficients and 95 percent confidence intervals from estimating the effect of a good manager's arrival on pricing-related outcomes. Panel (a) shows log sales of products with a price change, Panel (b) shows log sales of products without a price change, and Panel (c) shows the number of price changes. The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. Event time is measured in months relative to the arrival of a good manager, with the month immediately prior to arrival normalized to zero. All specifications include store and month fixed effects, and standard errors are clustered at the store level. The vertical line indicates the month of a good manager's arrival.

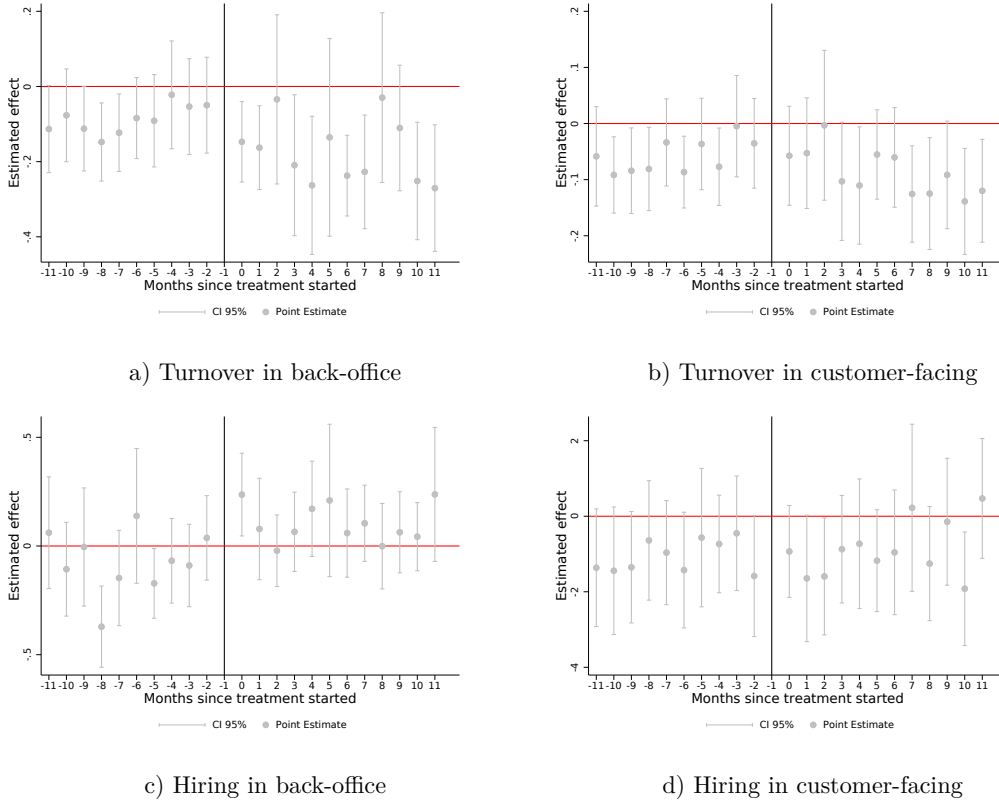
Table A.11: Absenteeism and Turnover

	(1)	(2)
	Absenteeism	Turnover
Treatment effect	1.093*** (0.378)	-0.788 (0.634)
Observations	9,007	9,007
R-squared	0.828	0.554
Treated Stores	81	81
Stores	220	220
Mean of Dv	6.835	4.742

Notes: Table A.11 shows the point estimates of absenteeism (measured as the number of unexcused worker absences in a month) and turnover (number of workers who quit in a month). The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. The treatment effect is defined as the arrival of a good manager to a bad store (i.e., a store in which the average fixed effect of incumbent managers is below that of the median store). Robust standard errors are in parentheses. * significant 10%, ** significant 5%, *** significant 1%.

A.2.2 Additional Evidence on Organizational Changes

Figure A.13: Effects on turnover, hiring, and moves in back-office and customer-facing departments



Notes: This figure shows event-study coefficients and 95 percent confidence intervals from estimating the effect of a good manager's arrival on turnover, hiring, and internal moves of managers in back-office and customer-facing departments. Panels report outcomes for manager turnover and hiring, separately by department type. The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. Event time is measured in months relative to the arrival of a good manager, with the month immediately prior to arrival normalized to zero. All specifications include store and month fixed effects, exclude the moving manager, and cluster standard errors at the store level. The vertical line indicates the month of a good manager's arrival.

Table A.12: Turnover, hiring, and moves in back-office and customer-facing departments

	(1) Turnover Managers (Back-office)	(2) Turnover Managers (Customer-facing)	(3) Total Hiring (Back-office)	(4) Total Hiring (Customer-facing)	(5) Moves Managers to Back-office	(6) Moves Managers to Customer-facing
Treatment Effect	-0.117*** (0.0335)	-0.0269 (0.0210)	0.188*** (0.0462)	-0.0842 (0.134)	0.00679 (0.0115)	-0.00154 (0.00476)
Observations	9,007	9,007	9,007	8,919	9,007	9,007
R-squared	0.483	0.235	0.277	0.564	0.061	0.076
Treated Stores	87	87	87	87	87	87
Stores	220	220	220	220	220	220
Mean of Dv	0.0754	0.0331	0.369	3.373	0.0169	0.00634
Relative Effect	-154.7%	-81.3%	51.0%	-2.5%	40.2%	-24.3%

Notes: Table A.12 shows the point estimates of manager turnover, hiring, and within-store moves in back-office and customer-facing departments. Back-office refers to departments whose main responsibilities are related to sales, pricing or inventory activities, while customer-facing departments are those with main responsibilities other than those mentioned above. *Turnover* is defined as the monthly number of manager departures per store. Similarly, *Hiring* is the number of workers hired in the month. As for *Moves*, this variable is constructed as all the movements that occur within the stores in a month, whether vertical or horizontal. The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. The treatment effect is defined as the arrival of a good manager to a bad store (i.e., a store in which the average fixed effect of incumbent managers is below that of the median store). Robust standard errors are in parentheses. * significant 10%, ** significant 5%, *** significant 1%.

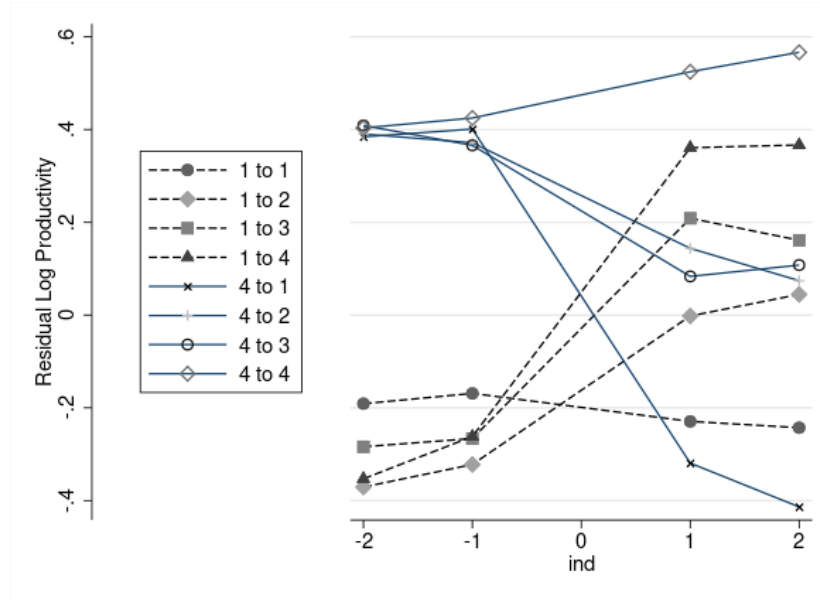
A.3 Additional Evidence from the AKM Decomposition

Table A.13: Explanatory power of AKM on log productivity

Covariance term	Fraction of $var(\omega)$
$Cov(\omega, x\beta)$	0.0916
$Cov(\omega, \theta)$	0.1205
$Cov(\omega, \psi)$	0.7183
$Cov(\omega, \varepsilon)$	0.0696

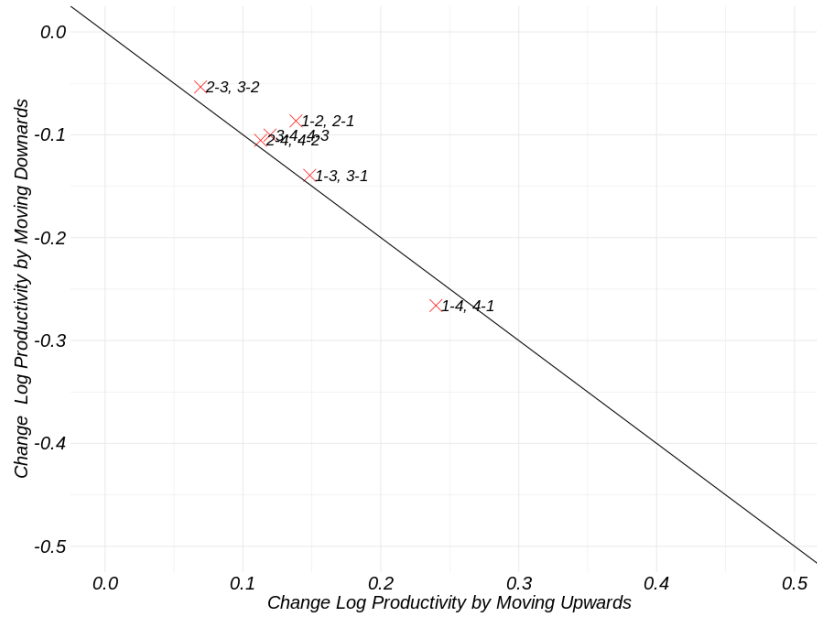
Table A.13 reports the explanatory power of each term in the AKM decomposition on the variance of productivity (ω). Each row refers to the covariance between log productivity and the covariates ($x\beta$), the manager fixed effects (θ), the store fixed effects (ψ), and the residuals of the model (ε).

Figure A.14: Event Study Around Movers - Log Productivity



Notes: Figure A.14 splits movers into quartiles based on average log productivity in their origin and destination stores. We label the origin quartile first, and the destination quartile second. So, for example, a manager leaving a fourth quartile (highest productivity) store for a first quartile (lowest productivity) store would be labeled '4 to 1.' We compute the productivity measure for each store over the entire sample period and plot the average residual of log productivity of movers on the vertical axis. We compute and plot the residual on the horizontal axis for the following periods: 3-4 months pre-move is period = -2; 1-2 months pre-move is period = -1; 1-2 months post-move is period = 1; 3-4 months post-move is period = 2. We create the residual by regressing monthly store log productivity on year and month fixed effects. We then predict the residuals and run the movers analysis. For parsimony, the graph plots only moves away from stores in the top and bottom quartiles.

Figure A.15: Symmetry Test - Log Productivity



Notes: Figure A.15 splits movers into quartiles based on average log productivity in their origin and destination stores. We label the origin quartile first, and the destination quartile second. So, for example, a manager leaving a fourth quartile (highest productivity) store for a first quartile (lowest productivity) store would be labeled ‘4-1.’ The points correspond to the average change in residual of log productivity of movers from stores in the origin quartile to the destination quartile plotted against movers going the opposite direction. So, for example, the point labeled “1-4, 4-1” represents the average change in log productivity for movers from stores in quartile 1 to stores in quartile 4, plotted against the average change for movers from stores in quartile 4 to stores in quartile 1. To calculate average productivity we restrict the sample to two months before and two-months after the move. We create the residual by regressing monthly store log productivity on year and month fixed effects. We then predict the residuals and run the movers analysis. The horizontal axis corresponds to “upward” moves and the vertical to “downward” moves. The line is drawn at 45-degrees. If upward and downward movement is symmetric all points should be on or close to the line.

A.4 Additional Robustness Results

Table A.14: Early- vs late-treated exogeneity check, time-FE residualized

	Coefficient	<i>t</i> -statistic	Mean	Obs
Sales (thousands USD)	22.2669	0.062	1547.575	87
Productivity (thousands USD)	-0.3084	-0.241	19.627	87
Log Sales (USD)	-0.0842	-0.286	21.771	87
Log Productivity (USD)	-0.0685	-0.831	18.015	87
Stockouts	-38.4786	-0.307	805.789	87
Inventory value (millions USD)	0.0168	0.221	0.423	87
Inventory value over sales	0.0179	0.766	0.253	87
Price changes	-88.5186	-0.032	12877.325	87
Products	3955.4476	0.367	48184.320	87
Total back-office workers	2.1902	0.321	23.834	87
Total customer-facing workers	12.0586	0.589	78.345	87
Total back-office managers	0.5518	0.460	5.244	87
Total customer-facing managers	1.1229	1.122	2.211	87
Total hiring (back-office)	0.0116	0.077	0.328	87
Total hiring (customer-facing)	1.4787	0.826	4.998	87
Hiring managers (back-office)	0.0084	0.767	0.007	87
Hiring managers (customer-facing)	0.0035	1.008	0.003	87
Moves managers to back-office	-0.0017	-0.104	0.008	87
Moves managers to customer-facing	0.0033	0.466	0.008	87

Notes: Each row reports the OLS coefficient from a cross-sectional regression of the store-level pre-treatment mean on an indicator equal to one if the store's first treatment month is at or below the median treatment month (February 2018) across ever-treated stores. Never-treated stores are excluded. Before collapsing, each outcome is residualized against month fixed effects, with the regression constant to ensure the adjusted series is centered at the mean. Treated stores contribute only pre-treatment periods to the collapse. Robust standard errors are in parentheses. * significant 10%, ** significant 5%, *** significant 1%. Monetary variables are in USD.

Table A.15: Alternative Definitions of “Treatment”

	Log Sales	Log Sales	Log Sales	Log Sales	Log Sales
<i>Panel A: arriving - existing management</i>					
1(Top 10%)	0.185*** (0.0437)				
1(Top 25%)		0.0891*** (0.027)			
1(Top 50%)			0.0338* (0.0192)		
1(Bottom 50%)				-0.0752*** (0.0185)	
Any Arrival					-0.0209 (0.0146)
Observations	9,007	9,007	9,007	8,684	8,684
R-squared	0.9833	0.9831	0.9830	0.9845	0.9843
Mean of Dep. Var.	22.02	22.02	22.02	22.03	22.03
<i>Panel B: arriving manager</i>					
1(Top 10%)	0.0274 (0.0903)				
1(Top 25%)		0.007 (0.0386)			
1(Top 50%)			-0.028 (0.0239)		
1(Bottom 50%)				-0.0228 (0.0175)	
Observations	9,007	9,007	9,007	9,007	
R-squared	0.9830	0.9830	0.9830	0.9830	
Mean of Dep. Var.	22.022	22.022	22.022	22.022	

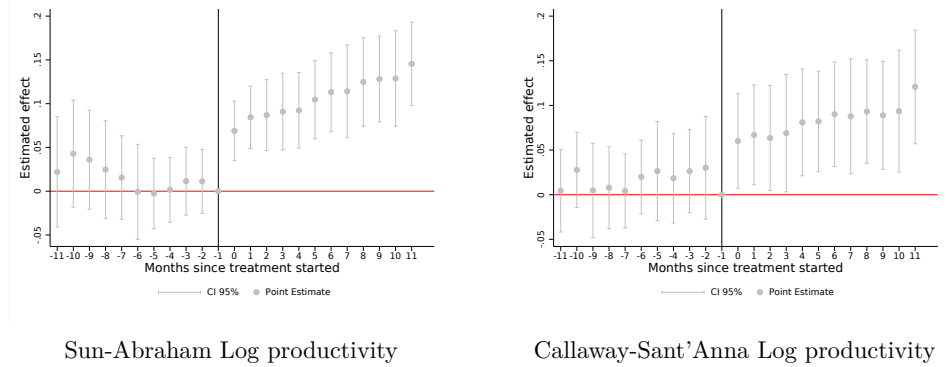
Table A.15 reports the effect of various alternatives definitions of treatment on log sales. $1(\cdot)$ is the indicator function. Panel A presents the difference between the (standardized) fixed effect (on log productivity) of the arriving manager and the mean of the (standardized) fixed effects of all managers who worked at the store immediately prior to arrival of the treating manager. We then sort these differences in descending order. Thus, a Top 10% difference represents cases in which the arriving manager is much better than the existing management. Each column corresponds to a subset of these sorted differences. Panel B ignores the fixed effects of the existing store management, sorting only the fixed effect of the arriving manager. Robust standard errors are in parentheses. * significant 10%, ** significant 5%, *** significant 1%.

Table A.16: AKM decomposition results and robustness

	Baseline	Andrews et al. (2008)	Leave-out Estimator
$\text{Var}(\theta)$	0.024	0.018	0.018
$\text{Var}(\psi)$	0.205	0.199	0.120
$\text{Cov}(\psi, \theta)$	-0.010	-0.005	-0.005
$\text{Corr}(\psi, \theta)$	-0.138	-0.081	-0.078

Table A.16 reports the baseline model which comes from the estimation of equation (1) following [Abowd et al. \(1999\)](#), the bias correction of [Andrews et al. \(2008\)](#) and the leave-out estimator from [Kline et al. \(2020\)](#). The data spans over January 2017 to June 2020 through 220 stores. For each model we compute the variance of the manager fixed effects $\text{Var}(\theta)$, the variance of the store fixed effects $\text{Var}(\psi)$, and the correlation $\text{Corr}(\psi, \theta)$ and covariance $\text{Cov}(\psi, \theta)$ of both types of fixed effects.

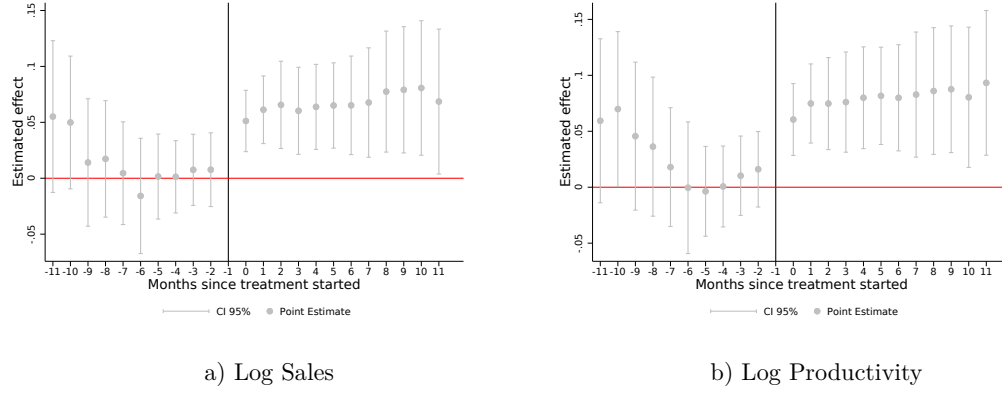
Figure A.16: Effects of a good manager on log productivity: Sun-Abraham and Callaway-Sant’Anna specifications



Notes: This figure shows the event-study coefficients and 95 percent confidence intervals from estimating the effect of a good manager’s arrival on log productivity using estimators that are robust to heterogeneous treatment effects under staggered adoption. Panel (a) reports the cohort-specific event-study estimates following [Sun and Abraham \(2020\)](#). Panel (b) reports group-time average treatment effects following [Callaway and Sant’Anna \(2020\)](#), translated into relative event time. The sample consists of store-monthly panel data for the 220 stores during our analysis period of January 2017 to June 2020. Event time is measured in months relative to the arrival of a good manager, with the month immediately prior to arrival normalized to zero. The vertical line represents the arrival of the good manager.

A.4.1 Not Yet Treated Stores

Figure A.17: Robustness: Not yet treated as controls



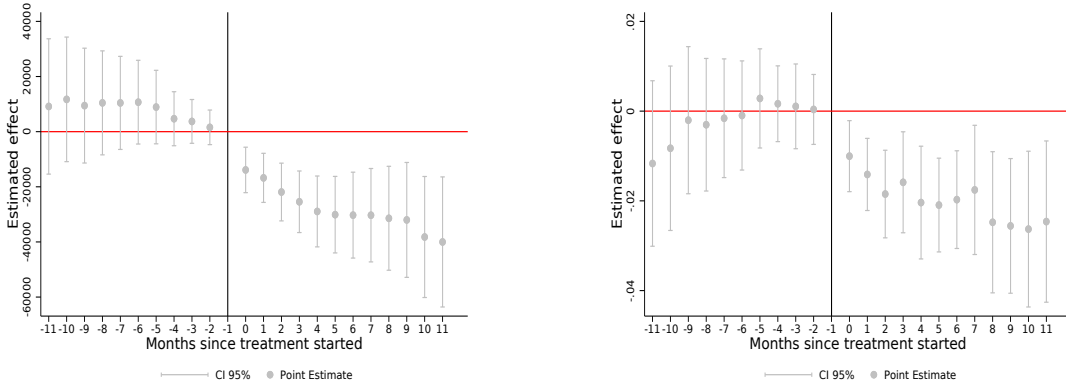
Notes: Each panel of this figure shows the event-study coefficients and 95 percent confidence intervals from estimating a good manager's arrival on Sales and Productivity. The sample consists of store-monthly panel data for the 87 stores during our analysis period of January 2017 to June 2020. The vertical line represents the arrival of a good manager.

Table A.17: Effects on stockouts and on-hand inventory value

	(1) Stockouts	(2) Inventory Value (USD)	(3) Inventory Value over Sales
Treatment effect	-61.87* (33.83)	-28,542*** (8,734)	-0.0158*** (0.00487)
Observations	3,562	3,562	3,562
R-squared	0.908	0.985	0.795
Treated Stores	87	87	87
Stores	87	87	87
Mean of Dv	754.6	401,042	0.250
Relative Effect	-8.2%	-7.1%	-6.3%

Notes: Table A.17 shows the point estimates of stockouts and on-hand inventory. A stockout is defined as a product whose number of items is equal to or below zero. The sample consists of store-monthly panel data for the 87 stores during our analysis period of January 2017 to June 2020. The treatment effect is defined as the arrival of a good manager to a bad store (i.e., a store in which the average fixed effect of incumbent managers is below that of the median store). Robust standard errors are in parentheses. * significant 10%, ** significant 5%, *** significant 1%.

Figure A.18: Value of inventory

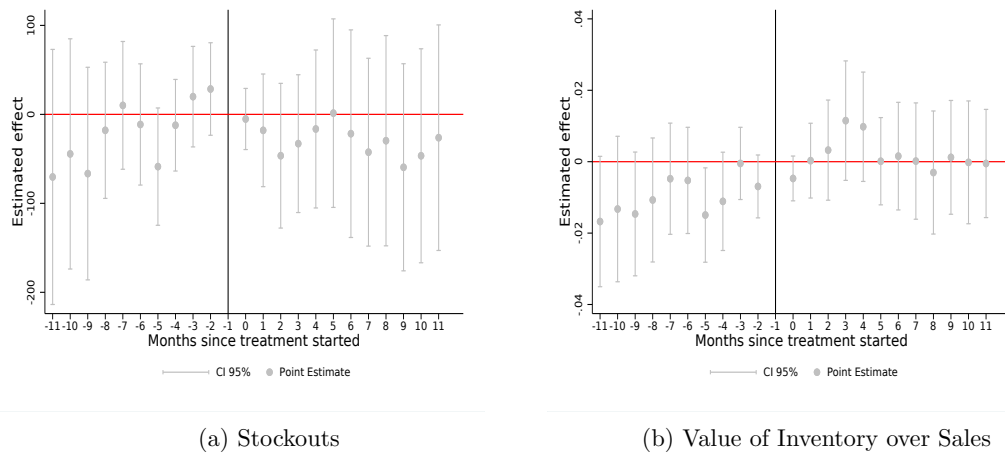


(a) Total value of inventory (USD)

(b) Value of inventory over sales

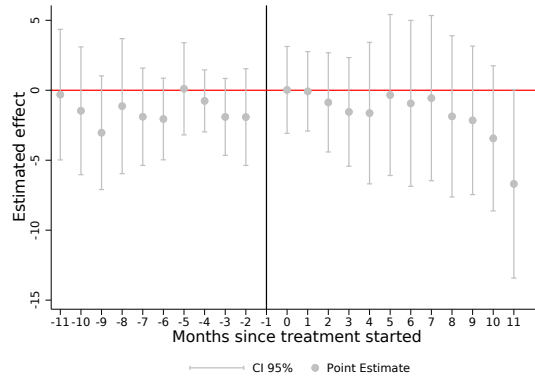
Notes: This figure shows event-study coefficients and 95 percent confidence intervals from estimating the effect of a good manager's arrival on (a) the total value of inventory (USD) and (b) the value of inventory divided by total sales. The sample consists of store-monthly panel data for the 87 stores during our analysis period of January 2017 to June 2020. Event time is measured in months relative to the arrival of a good manager, with the month immediately prior to arrival normalized to zero. All specifications include store and month fixed effects, and standard errors are clustered at the store level. The vertical line indicates the month of a good manager's arrival.

Figure A.19: Effects of Manager Departure on Stockouts and Value of Inventory on Hand



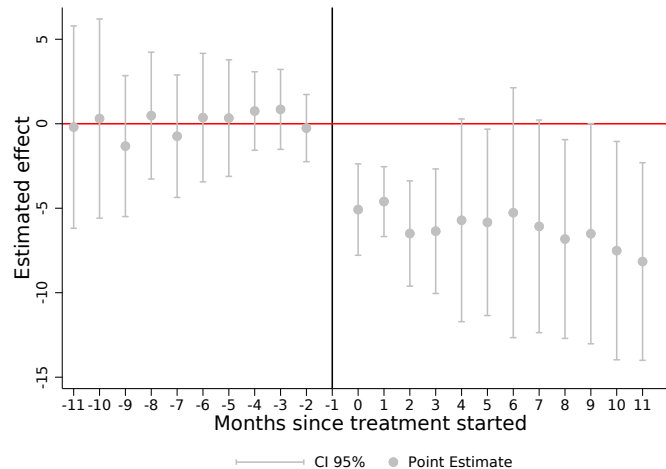
Notes: This figure shows the event-study coefficients and 95 percent confidence intervals from estimating a good manager's *departure* on the number of stockouts and the value of inventory in a store (in USD). The sample consists of store-monthly panel data for the 87 stores during our analysis period of January 2017 to June 2020. The vertical line represents the departure of a good manager.

Figure A.20: Number of Workers



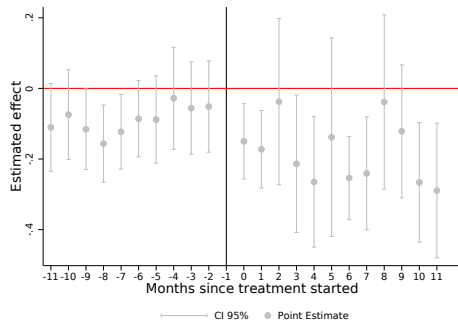
Notes: This figure shows event-study coefficients and 95 percent confidence intervals from estimating the effect of a good manager's arrival on the total number of workers in a store. The sample consists of store-monthly panel data for the 87 stores during our analysis period of January 2017 to June 2020. Event time is measured in months relative to the arrival of a good manager, with the month immediately prior to arrival normalized to zero. All specifications include store and month fixed effects, and standard errors are clustered at the store level. The vertical line indicates the month of a good manager's arrival.

Figure A.21: Effect on number of trainings

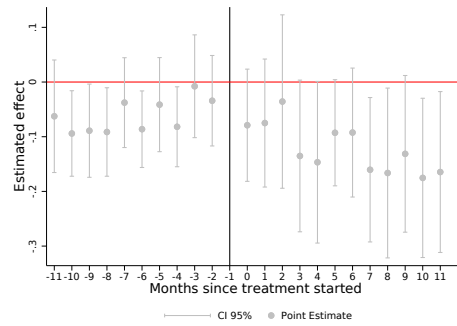


This figure shows event-study coefficients and 95 percent confidence intervals from estimating the effect of a good manager's arrival on the number of trainings conducted in a store each month. The sample consists of store-monthly panel data for the 87 stores during our analysis period of January 2017 to June 2020. Event time is measured in months relative to the arrival of a good manager, with the month immediately prior to arrival normalized to zero. All specifications include store and month fixed effects, and standard errors are clustered at the store level. The vertical line indicates the month of a good manager's arrival.

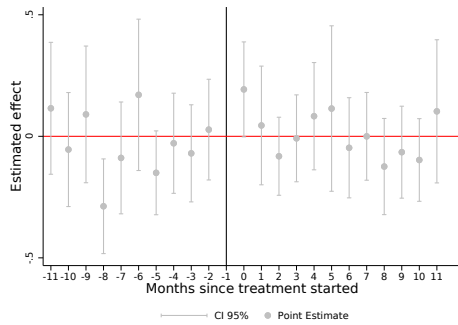
Figure A.22: Effects on turnover, hiring, and moves in back-office and customer-facing departments



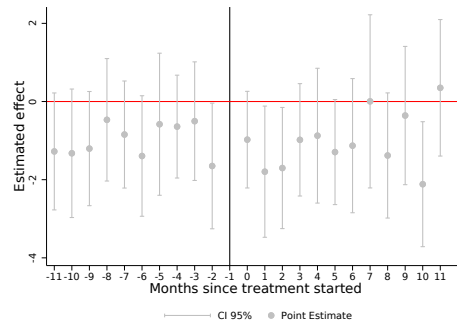
a) Turnover in back-office



b) Turnover in customer-facing



c) Hiring in back-office



d) Hiring in customer-facing

Notes: This figure shows event-study coefficients and 95 percent confidence intervals from estimating the effect of a good manager's arrival on turnover, hiring, and internal moves of managers in back-office and customer-facing departments. Panels report outcomes for manager turnover and hiring, separately by department type. The sample consists of store-monthly panel data for the 87 stores during our analysis period of January 2017 to June 2020. Event time is measured in months relative to the arrival of a good manager, with the month immediately prior to arrival normalized to zero. All specifications include store and month fixed effects, exclude the moving manager, and cluster standard errors at the store level. The vertical line indicates the month of a good manager's arrival.